UNIVERSITI TEKNOLOGI MALAYSIA

PROJECT

TASK 6A: MAKING A GROUP REPORT

Course : SECR1213 NETWORK COMMUNICATION

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Section : 02

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ABSTRACT

This task 6A reports compile all of our group progress in case study of FC to build a new 2 –storey building that grows and moves forward into the future, so does the need for better equipment that will allow for this growth.This report sum ups our plans starting from task 1 untill task 5 starting on proposing a well-equipped labs—two general-purpose labs, a Cisco Network lab for networking education, and an Embedded lab for IoT and digital devices—each sized at 14m x 10m with 30 workstations. We have also incorporated a hybrid classroom, video conferencing room, and a student lounge that will serve as a starting point for us to design a networking architecture that has fast connectivity for study and relaxation.

We also did preliminary analysis, especially on current and future requirements and devices to be used on Task 2.This task requires us to generate numbers of question and ask the user or to research (the Internet, white papers, journals, etc.) with regards to the requirements and other information that is necessary to develop a network plan based on the case study.We then continue on find out what devices (network and end-user devices) that we will need to achieve the objective of fulfilling the requirements and needs of the organization.We also made some reflections especially on the prices of networking devices available on market in this task 3.

Moving on Task 4, we have put all the infrastructure and devices we chose into the floor plan that we have created earlier.This task started firstly by creating the Network Distribution Diagram ,inserting the cable information on all floor diagram, determining the number of connections,patch cords and switch ports needed and ended the task by identifies the cable type and length.Then we continue on the crucial part in Task 5 that is IP addressing in making sure that every host can connect to the network without conflict of addresses.We received 10.16.0.0/12 the subnetwork from the Network Address assigned to my group. We completed and detailed the workings of our subnetting and IP assignment to each lab and room.

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INTRODUCTION

The aim of this project is to come up with a network architecture for the two-story building belonging to the Faculty of Computing. The objective is to establish a technological facility that maximizes learning and collaboration throughout the process of learning. The design of the building includes general-purpose labs, a Cisco networking lab for hands-on networking experience, an IoT-capable embedded lab, hybrid classrooms for both physical and virtual learning, video conferencing rooms, and a student lounge equipped with reliable connectivity.

Moving on to the next part, the project scope includes planning and designing the functionality as well as the efficiency of the network facilities. This involves the position and placement of the equipment and workstation in addition to selecting the network devices such routers, switches, cables by taking account their performance and durability. Our topmost concern is to ensure that there is a strong network infrastructure with powerful connectivity that guarantees fast data transmission and seamless communication, including fulfilling all the needs for scalability and security.

The main goals of this project are also to provide fast and reliable connectivity that satisfies both the academic and operational parts of the faculty, students, and staff. The project itself will feature smooth communication, allow for future scalability and ensure great network security. We assume that the resources are enough to provide a consistent network usage throughout the building, which will enable efficiency and consistency in the performance of the network.

This initial strategy of ours will supply the network efficiency, and future ready for developments.he project emphasizes performance, scalability, and security. This helps it to meet today's academic and administrative needs while preparing for future technology advancements.

PROJECT BACKGROUND AND AN OVERVIEW OF THE CLIENT'S CURRENT STATUS AND ISSUE

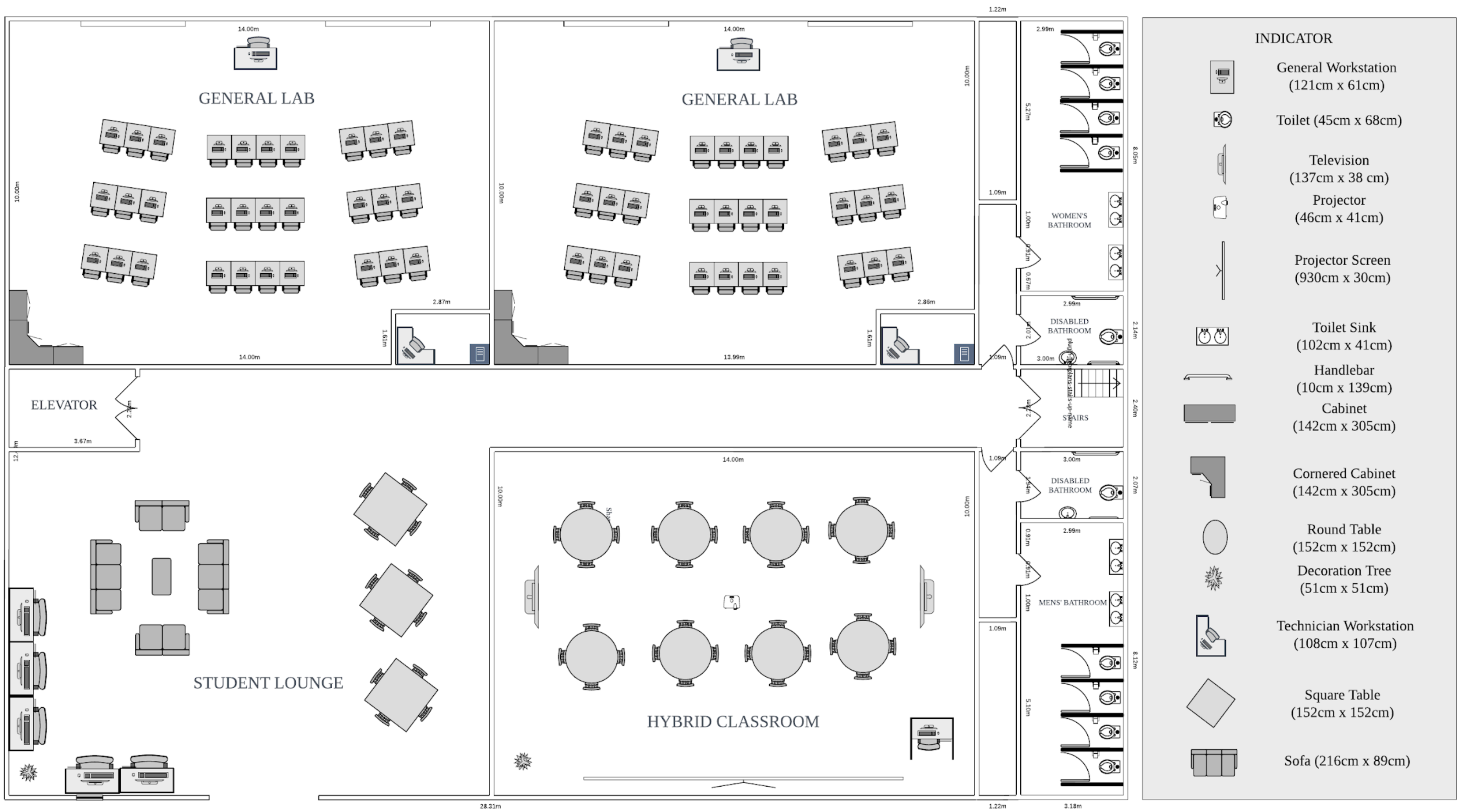
The Faculty of Computing (FC) is growing rapidly, and, as predicted, both student enrollment and academic personnel are expected to increase over the next four years. As demand increases, FC is planning a new two-story building that would meet the growing needs of students, staff, and academics better. This building will involve various special-purpose labs, one hybrid classroom, a video conferencing room, a student lounge, among others, which require an efficient and reliable network infrastructure.

Currently, FC's network infrastructure is not in a position to accommodate the expansion that is about to take place, especially with the expectations brought forward by 4IR, which require high-speed internet and modern networking. Currently, the network infrastructure cannot support emerging demands for speed, higher data throughput, and scalability that academics and students need. Besides, the network should be secure, manageable, and cost-effective.

FC has to address several challenges: network infrastructure that scales with institutional growth, while being secured from many kinds of cyber threats, including DoS; high-performance networking needs for many special labs including the Cisco Networking Lab and Embedded Lab; assurance of wireless coverage within the building.

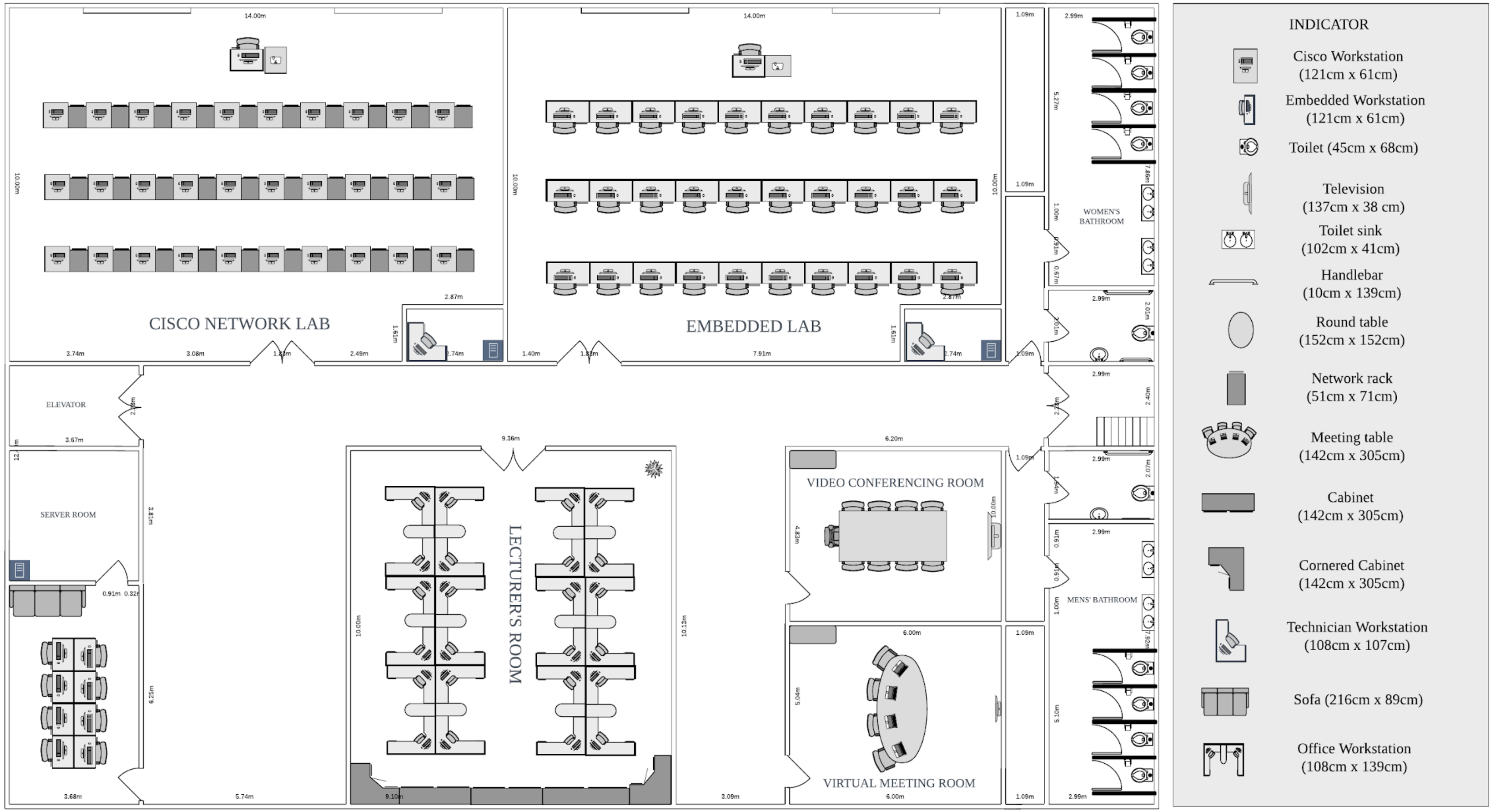
TASK 1: PROJECT SETUP

1. 1 GROUND FLOOR - FLOOR PLAN



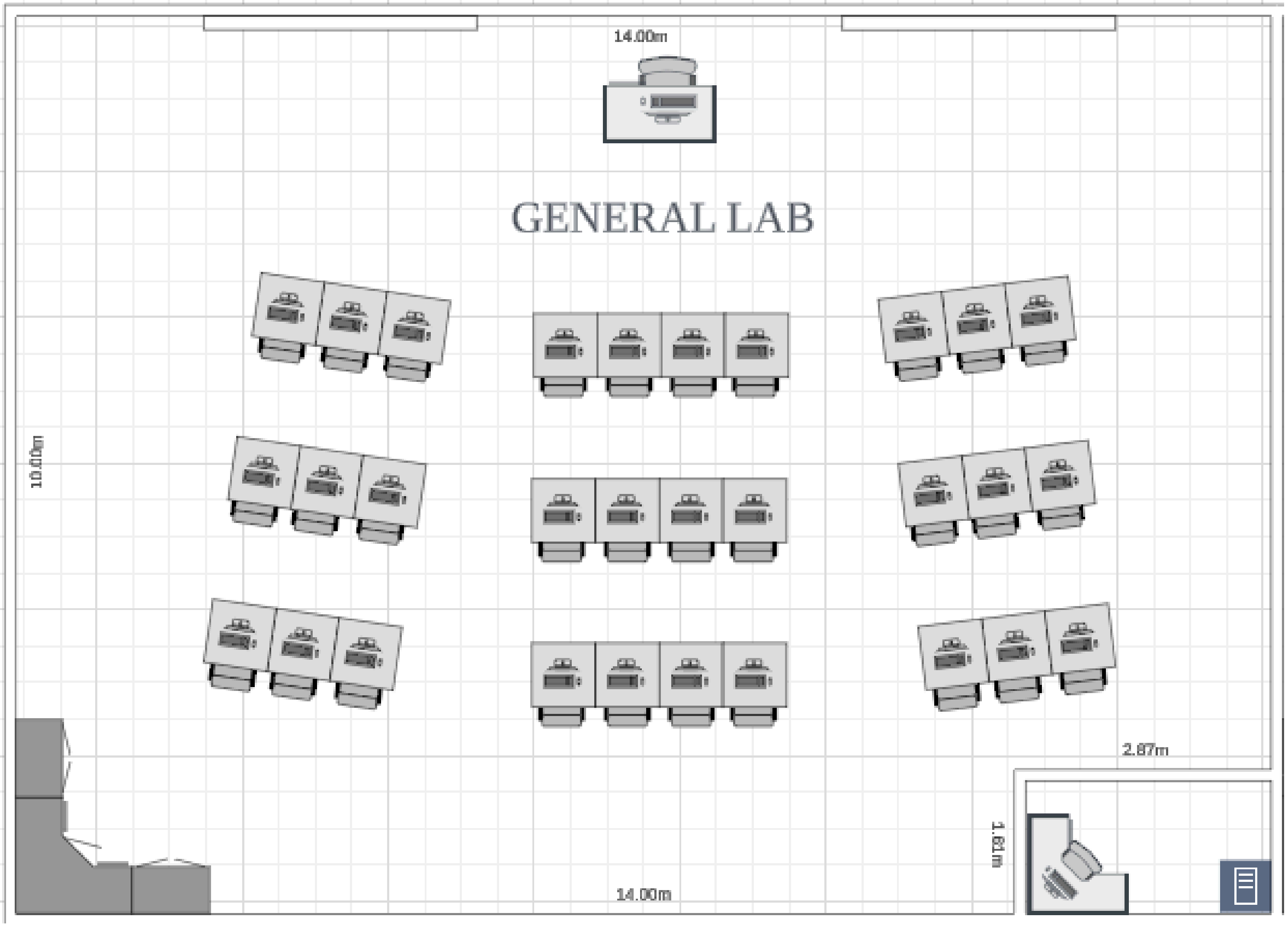
Scale, 1cm : 1m

1.2 FIRST FLOOR - FLOOR PLAN

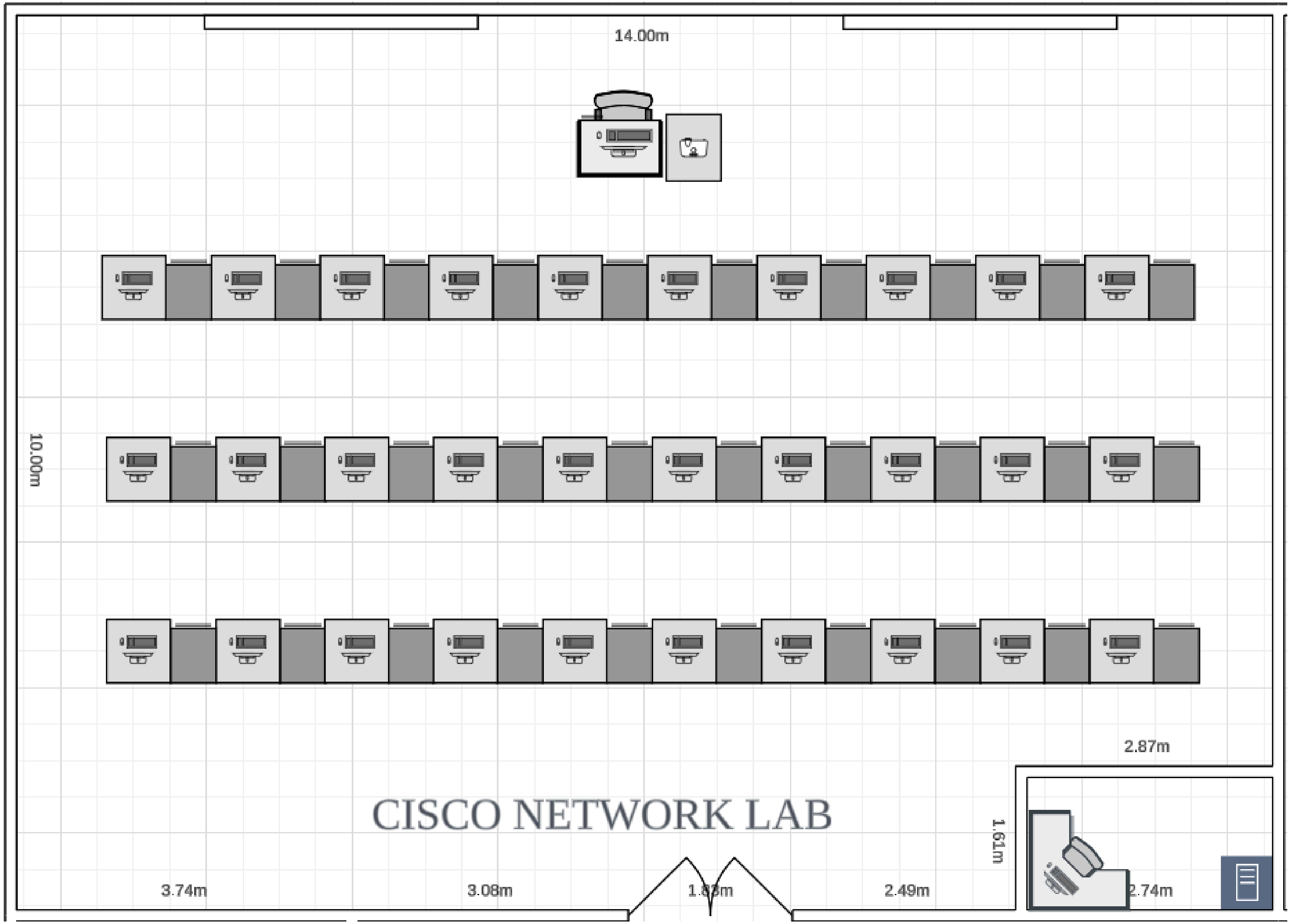


Scale, 1cm: 1m

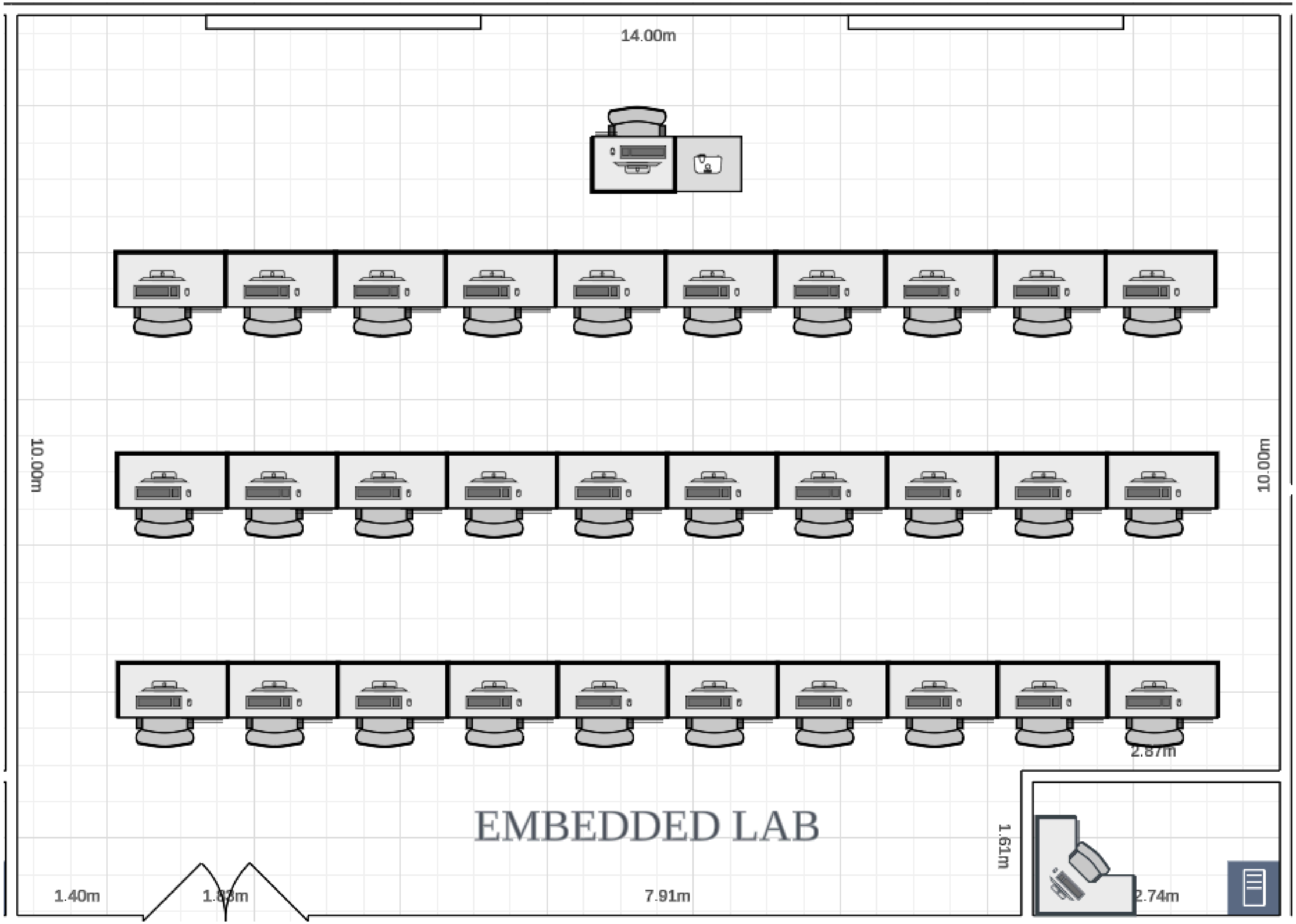
1.3 GENERAL PURPOSE LAB - FLOOR PLAN



1.4 CISCO NETWORK LAB - FLOOR PLAN



1.5 EMBEDDED LAB - FLOOR PLAN



TASK 2: INITIAL DESIGN - PRELIMINARY ANALYSIS

2.1 QUESTIONS :

1. **What are the technical and spatial requirements to ensure the hybrid classroom has the best performance setup for in-person and remote teaching?**

Integrating technological and spatial components that support smooth in-person and distant learning experiences is crucial for optimizing hybrid classrooms[1]. High-quality audio-visual tools, like cameras and microphones, and dependable, fast internet access are essential for facilitating active communication between students who are enrolled remotely and those who are there. A more seamless experience is also made possible by sophisticated, adaptable technology that dynamically changes audio and visual settings. This helps to maintain a uniform learning environment in both settings and lessens the burden on teachers to handle technical problems.[1]

Apart from technical requirements, the classroom's flexibility and participation are greatly impacted by spatial design[2]. For instance, highlight how adaptable, modular furniture arrangements and well-placed screens and microphones provide a setup that motivates both in-person and remote students to participate completely. This spatial arrangement enhances the overall hybrid experience by supporting students' variety of learning demands in addition to facilitating collaborative activities.[2]

Both technical and spatial components are essential for a successful hybrid learning environment. Remote students can participate completely alongside in-person participants thanks to dynamic audio-visual equipment, high-quality cameras, and dependable, fast internet [1]. Furthermore, adaptable, modular furniture and well-placed screens and microphones foster an inclusive atmosphere that accommodates different teaching philosophies and promotes student cooperation [2].

2. **What is the anticipated bandwidth requirement for the 4 labs and student lounge to ensure high-speed internet access?**

After analyzing the network requirements for each area in the case study, it has been determined that each lab will need a minimum of 300 Mbps to support an efficient workflow. Each lab can accommodate around 30 computers, with each computer requiring a bandwidth of at least 10-25 Mbps to handle tasks like online browsing, downloading resources, and accessing digital learning materials [3]. This level of bandwidth ensures that users experience minimal lag, even during peak hours when multiple computers are in use simultaneously.

Furthermore, for an optimal experience, the network’s minimum latency should be around 30-40 milliseconds. This low latency is crucial for smooth operation, especially when users are working with real-time applications, engaging in virtual labs, or attending online classes where immediate response times are needed to enhance learning and productivity.

For the student lounge, which is primarily used for casual browsing, social media, or light streaming activities, the bandwidth requirements are lower than in the labs. Here, each user should have a bandwidth of around 10-15 Mbps to ensure smooth access to these applications without excessive buffering or delays [4]. Assuming that around 30-40 students will be using the lounge network simultaneously, the overall required bandwidth for the lounge is estimated to be around 300-600 Mbps. This allocation allows for multiple users to browse, stream, or access social media while ensuring a satisfactory user experience.

3. **What networking devices are required for the Cisco Network Lab to support teaching, and how will they align with future industry needs?**

In a standard Cisco Network Lab, the key networking devices typically consist of routers, switches, access points, and ASA (Adaptive Security Appliance) firewalls [5]. These components establish the fundamental framework of the lab setup, delivering the essential functionality needed for various networking tasks. Additionally, a highly adaptable device that can be incorporated into the network lab is the Raspberry Pi. This compact yet powerful computer can fulfill a range of roles when linked to Cisco devices. For example, the Raspberry Pi can function as an access node, enabling connections between learners and Cisco devices, as well as acting as a link between the end computers and the network infrastructure [6]. In addition to its role as an access node, the Raspberry Pi can serve multiple other purposes. It can work as a lab management station, allowing users to oversee and organize the devices in the lab. Moreover, it can act as a network monitoring device, enabling students to track and analyze traffic and performance across the network. Furthermore, the Raspberry Pi can be set up to run specific applications, providing additional versatility in its application within the lab environment [7].

The addition of these devices in a Cisco Network Lab arrangement equips students with practical, hands-on experience. Engaging with routers, switches, firewalls, and the Raspberry Pi offers students the opportunity to deepen their understanding of networking and communication principles. These practical experiences are essential for developing the skills necessary to meet the challenges of the rapidly changing tech industry. Thus, the lab not only aids students in grasping theoretical concepts but also prepares them for real-world challenges, enhancing their proficiency in working with contemporary network infrastructure and devices.

4. **How will the video conferencing room be integrated into the overall network to support virtual project meetings with stable, high-quality connectivity?**

The collaborative video-conferencing network enables end users to participate in an encrypted audio and video conference, which is a virtual space that includes facilities for collaborative unified communications such as instant messaging, presence, contact management, videoconference participant management, share screen, and privileged virtual spaces [8].

When selecting a virtual meeting platform, high-quality video and audio are essential [9]. The bandwidth for videoconferencing and streaming, numerous point-of-sale transactions and frequent file-sharing is about 75 mbps for 5-10 users [3]. It’s hard to have a productive meeting if the sound cuts out or the video is blurry. To ensure smooth performance, performance tests must be run when setting up the software and before each meeting [9].

During an in-person meeting, it's simple to distribute a document or show a presentation. The virtual meeting room software must be convenient for presentation. Participants should be able to easily share screens and collaborate on all main sorts of document files [9].

5. **How can the network be optimized to support cloud-based applications and services that the Faculty of Computing may adopt in the future?**

There are many approaches to optimising network performance, each of which contributes to a more efficient and efficient network system. The first way is data-driven optimisation, which uses monitoring tools to detect and resolve network bottlenecks[10]. This software continuously monitors network performance and collects vital data on potential issues such as latency, packet loss, and congestion. Analysing this data allows network administrators to identify the main reasons of performance decline and deploy specific approaches to solve them. Over time, this process results in a more flexible and efficient network, as each fixed issue provides insights that prevents future occurrences, allowing for continuous performance improvement.

Network optimisation can also be done by effectively managing network protocols, particularly those aimed for congestion control and packet size limitation. Protocols such as Transmission management Protocol (TCP) provide congestion management features that dynamically modify the flow of data to keep the network from getting congested. Furthermore, packet size management techniques help optimize how data is split down and transferred across the network, ensuring that packets are neither too large, which can cause delays, nor too little, which can cause inefficiencies.

Furthermore, bandwidth optimisation benefits significantly to network speed. This strategy involves improving the capacity of data flow across the network using techniques such as data compression and caching. Compression minimises the amount of data that must be delivered, allowing for better use of available bandwidth. Compressing data allows more information to be transferred in less time, significantly enhancing network speed and throughput. Caching, on the other hand, keeps frequently accessed data locally, either on a server or on a device, eliminating the need to retrieve the same data from remote sources many times. This not only reduces network traffic but also speeds up data access for users, resulting in a significant boost in performance.

6. **How will the network handle high-traffic scenarios, such as simultaneous lab sessions or large-scale virtual meetings in the video conferencing room?**

Networks manage high-traffic conditions using a variety of techniques, with load balancing being an essential one. Load balancing effectively distributes traffic over multiple servers or network connections, preventing any single resource from getting overloaded. This optimizes resource utilization and improves user experience by lowering latency and preventing overloads [11]. For example, in a high-traffic virtual lab or conference, load balancing distributes participant requests across servers, ensuring that all users experience smooth and responsive performance.

The second one is VLANs and subnets. Using VLANs and subnet in high-traffic scenarios, like simultaneous lab sessions or large-scale virtual meetings, can significantly enhance network efficiency by segmenting traffic and optimizing bandwidth [2]. VLANs and subnets group endpoints based on function, ensuring that most traffic remains within each segment, thus reducing unnecessary cross-network data transfer. For instance, lab devices would primarily interact within the lab VLAN, minimizing their impact on other segments. In the video conferencing VLAN, Quality of Service (QoS) rules can prioritize video and audio packets, reserving bandwidth for smoother communication, even during network-heavy events [11]. Additionally, VLANs and subnets add layers of security and redundancy by enabling specific access controls for each segment, preventing disruptions across VLANs and ensuring critical resources are protected. This strategic segmentation reduces latency and helps maintain consistent performance, even in high-traffic conditions [12].

Lastly, the importance of high-quality hardware cannot be overstated. Investing in reliable routers, switches, and other networking infrastructure components provides a solid foundation that can handle increased traffic loads effectively [11]. This is important in scenarios such as simultaneous lab sessions or a heavy-use virtual meeting. In these cases, numerous users rely on consistent, high-speed connectivity to access shared resources, simulations, or video conferencing tools.

7. What disaster recovery plan will be in place to protect against data loss and network outages?

In formulating a thorough disaster recovery strategy for the new Faculty of Computing building, it is crucial to establish strong measures that guarantee business continuity and safeguard data. The strategy should encompass both the physical infrastructure, demonstrated by the careful positioning of the server room on the first floor to minimize flood hazards, as well as digital protections against various types of data loss and network failures.

An essential element of the disaster recovery plan involves setting up consistent data backups, which will protect against data loss by saving copies of important files, applications, and databases in secure, off-site, or cloud storage[13]. Utilizing cloud-based services can help minimize recovery times by allowing quick retrieval of data even when the local network is compromised while also allowing the data to be process, save, and view anytime[13] . It is advisable to arrange regular, automated backups on a daily or weekly basis, depending on the sensitivity and usage of the data, while employing incremental backups to prevent storage excess and ensure information remains current.

Having redundant network infrastructure is a crucial element of Disaster Recovery planning. By establishing network redundancy, an organization can guarantee that essential network operations have alternative resources ready in case of equipment malfunctions or network disruptions [14]. This can be accomplished by setting up secondary network routes, duplicate servers, and load balancers that manage traffic if a primary connection. Furthermore, incorporating backup power sources, such as uninterruptible power supplies (UPS) and backup generators, will enhance protection against power interruptions and assist in maintaining network reliability[14].

In summary, the disaster recovery strategy for this structure should prioritize consistent data backups, network redundancy, real-time data replication, and periodic testing. Together, these elements, along with frequent revisions to the disaster recovery plan, will create a robust framework to safeguard the building's network against data loss and maintain business continuity.

8. **What measures will be put in place to ensure the new building’s network is secure against cyber threats such as Internet Worms, denial-of-service attacks, and e-business application attacks?**

To protect the new faculty building's network from cyber threats such as Internet worms, denial-of-service (DoS) attacks, and e-business application vulnerabilities, a comprehensive defense strategy that includes strong security practices and technologies is vital. This building, which comprises labs, classrooms, and video conferencing facilities, will accommodate a variety of devices and network users, thereby increasing its susceptibility to cyber threats. Consequently, it is essential to develop a secure and resilient network architecture to safeguard sensitive information and ensure continuous network availability.

The initial defense against cyber threats involves setting up robust perimeter security. A firewall should be positioned at the network's access points to oversee and regulate incoming and outgoing traffic . Firewalls are effective in blocking unauthorized access and preventing the proliferation of worms, which use open network ports to infect systems [15]. Furthermore, implementing an intrusion detection and prevention system (IDPS) alongside the firewall can aid in recognizing and responding to malicious activities within the network by analyzing traffic for known attack patterns [16]. An IDPS can also counter denial-of-service attacks by detecting and halting unusual traffic surges before they exhaust network resources, thereby ensuring service continuity.

An additional essential step is dividing the network into zones, which is crucial due to the varied areas within the facility [17]. By segmenting the network, important resources like the server room and technician's office are kept separate from common access regions, minimizing the consequences of a possible security breach. For example, implementing VLANs (Virtual Local Area Networks) facilitates the secure separation of lab networks from the administrative network, stopping unauthorized lateral movement across the system if one section is compromised.

9. **What monitoring and maintenance tools will be implemented to proactively detect and resolve network issues before they affect users?**

For the new Faculty of Computing building, which features specialized areas such as the Cisco Network Lab and rooms for video conferencing, it is essential to establish a thorough Network Management System (NMS) [18]. The infrastructure for monitoring should include several fundamental components.

To begin with, a centralized monitoring platform, utilizing tools like SolarWinds Network Performance Monitor or Nagios XI, ought to be set up in the Technician's Office. These tools can deliver real-time insights into various network performance metrics, such as bandwidth usage, latency, and packet loss [16]. The monitoring system must utilize a Simple Network Management Protocol (SNMP) to gather data from network devices throughout all labs and classrooms.

Furthermore, predictive analytics along with machine learning algorithms can be utilized to identify anomalies prior to affecting users. Research has indicated that using AI-powered monitoring tools can cut network downtime by as much as 50% by enabling the early identification of potential problems [19]. Additionally, the system ought to incorporate automated ticketing integration to enhance the resolution process when issues arise.

10. **What type of cable is suitable for to setup a networking environment among computers based on the specification?**

There are three main cables that can be used for the networking setup which are coaxial cable, twisted pair network cable and fibre optic cable. For general use in labs such as the general purpose lab that does not need a very high internet speed, a Cat 6 shielded twisted pair (STP) network cable is recommended as the max bandwidth and max transmission speed is 250Mhz and 10Gbps respectively [20]. In addition, shielded copper wire provides protection against protection from electromagnetic inteference and radio frequency interference for faster data transfer rate. For Cisco network lab and embedded lab however, they require a significantly higher internet speed for uninterrupted learning session. A Cat 6a shielded twisted pair network cable is recommended as the max bandwidth and max transmission speed is 500Mhz and 10Gbps respectively[20]. Categories below 5e are not recommended as it is deemed to be outdated and is not compatible with modern LAN network [21] The main reason why STP cables are chosen is because cheaper than other cables are easier to setup. Compared to fibre optic cable, evethough it has superior transmission speed and bandwidth, it is considerably more expensive and is less durable than STP cables [22]. Coaxial cable on the other hand is not suggested to be used because it is prone to signal leakages and has poor long distance transmission [23].

2.2 FEASIBILITY ASSESSMENT

Technical Feasibility

The project is shown to be technically feasible. The network requirements, equipment and devices needed to align with the faculty's goals after careful considerations such as network scalability and wireless connectivity. The inclusion of emerging technologies such as IoT devices, digital sensors, artificial intelligence and cloud computing demonstrates future-ready network infrastructure.

Economic Feasibility

In terms of economics, the project seems to also be feasible. The devices used are carefully chosen and researched so that it is cost-effective without compromising on performance and quality standards. Proper resource management methods suggested by the research also indicate positive economic feasibility.

Operational Feasibility

Operationally, the project seems to be feasible. The network architecture and design equipped with prevention, backup and recovery plans demonstrate the network architecture is ready to operate even if there are complications. Network management strategies ensure the network can support high amounts of users a a given time without sacrificing performance.

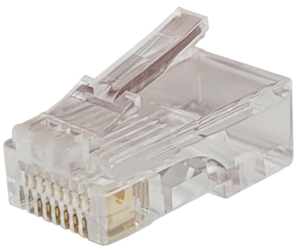
TASK 3: CHOOSING THE APPROPRIATE LAN DEVICES

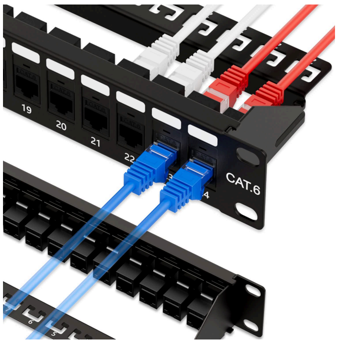
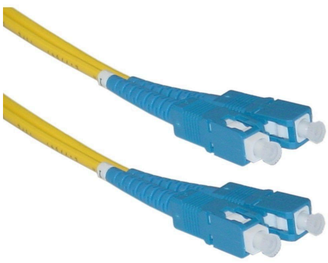
There are many different types of network devices like routers, switches, patch panels, wireless devices, and cables. These devices allow hardware on a computer network to communicate and interact with each other to help manage and direct data flow in a network. They ensure efficient communication between connected devices by controlling data transfer, boosting signals, and linking different networks. The devices that are planned to be used will be detailed in this report.

3.1 TABLE OF DEVICES

Device	Description	Price(RM/unit)	Quantity	Total(RM)
Router: FS SG-5105, Multi-WAN Security Gateway 	● Recommended Concurrent Users : 500 ● Fixed Ports: 8x 1000BASE-T, 1x SFP, 1x SFP+ ● WAN Ports: 9x 1 Gigabit, 1x 10 Gigabit WAN ● Flash Memory: 8 GB ● RAM: 2 GB	3747.00	3	11,241.00
Wireless Access Point: FAP-431F	● Support Bluetooth Low	3,057.00	8	24,456.00

	Energy Radio ● Simultaneous SSIDs up to 8 per client serving radio (7 if background scanning is enabled) ● Ports: 1x 100/1000/2500 Base-T RJ45, 1x 10/100/1000 Base-T RJ45, 1x Type A USB, 1x RS-232 RJ45 Serial Port ● PoE Port : 802.3at PoE default and 1 port powered by 802.3at or 2 ports powered by 802.3af			
Modem: FortiWiFi-81F-2R-3G4G-PoE 	● Internal Storage: 1x 128 GB SSD 1x 128 GB SSD ● Concurrent Sessions (TCP): 1.5 Million ● Ports: GE RJ45/SFP Shared Media Pairs x2, GE RJ45 POE/+ Ports x6, GE RJ45 POE/+	18,670.00	1	18,670.00

	FortiLink Ports (Default) x2 ● Support firewall ● SIM Slots: (Nano SIM) x2 ● Wi-Fi: Dual WiFi Radio (5 GHz, 2.4 GHz) with supported WiFi standard 802.11a/b/g/n/ac/ax + 1 Scanning Radio			
Connector: MP88X-1000 	● Number of Positions/Contacts: 8p8c (RJ45, Ethernet) ● Shielding: Unshielded ● Contact Finish: Gold ● Contact Material: Copper Alloy	2.42	1000	1,101.56
Ethernet cable : LSZH Indoor IEC60332-3-24Category 7A Enhanced Cable 	● ANSI/TIA Category:7A ● Delay:12ns/100 m ● Switching capacity:216Gbps ● Forwarding rate:162 Mpps ● Power supply:100-240	15.09 /m	2500	37,725.00

	VAC, 50/60 Hz			
Switch : X1052P - Dell Networking X Series Switches 	● Ports : 48 ● Switching capacity : 176 Gbps ● Forwarding rate :162 Mpps ● Power supply :100-240 VAC, 50/60 Hz	2,126.00	9	19,134.00
Patch Panel : 	● Connector Type - RJ45 ● Cable Type - Ethernet	223.47	20	4,469.4
Fiber Optic Cable : 	● SC-SC-10-Meter-Singlemode-Fiber-Optic-Cable	50.56/10m	350	1,769.6
Router for Cisco Networking Lab: Cisco C921-4P	● WAN interfaces: 2 ports Gigabit Ethernet (GE) ● LAN interfaces:	4,461.57	8	35,692.56

	4-port GE managed switch ● Integrated USB 2.0: Yes ● DRAM: Default 1 GB ● Flash memory: 2 GB			
Switch for Cisco Networking Lab: Cisco WS-C2960C-8TC-S 	● Ports: 8 x 10/100 Fast Ethernet ● Forwarding Bandwidth: 10 Gbps ● RAM: 128 MB ● Flash Memory: 64 MB	3,275.21	8	29,801.68

Table 1: List of Devices

3.2 REFLECTION

Are you surprised by the prices? How were you surprised?

Yes ,I'm quite surprised about the prices of the networking devices. Although it depends on the device's characteristics and circumstances, I admit that I am a little surprised by the cost of networking devices. Because of their advanced functionality, scalability, and durability, high-performance networking devices like enterprise-level switches, routers, and cables frequently have higher prices. For example, it makes sense that switches with Power over Ethernet (PoE) capabilities or those with larger bandwidths, such as 10G or 40G, are more costly.

The price difference between enterprise-grade and consumer-grade devices can still be shocking though. The price is heavily influenced by the price of advanced security measures, brand repute, and proprietary technologies. On the other hand, low-cost devices with restricted features might be acceptable for personal use or smaller networks.Overall, while the prices can seem steep initially, they reflect the importance of investing in reliable, efficient, and future-proof networking equipment, particularly for institutions or businesses where downtime or poor performance could result in substantial losses.

Have you ever considered cost as a factor for choosing networking devices?

On one hand, we considered that cost as a factor for choosing networking devices. This is mainly due to the budget constraint given for the project to operate therefore we try to find the best-suited devices that will work efficiently for our project yet does not cost so much. Networking devices like routers, switches and firewalls come in a wide range of prices with a variety of drawbacks and upper hands making it crucial to consider deeply before choosing one for this project. However, cost can also be an unexpected factor that can often lead to over-expenditure from the initial budget to match up the compatibility of the current technology such as fiber optic cable for a higher and more consistent internet connection that will replace coax, twisted pair etc. On the other hand, putting a little more money upfront into dependable and sturdy devices can often lead to savings down the road, which makes the connection between cost and value a bit more interesting than it first seems

What are the major differences between the same devices from different brands?

Several comparisons were made between the same devices from different brands to determine the best one to be used for this project. The comparisons were made in a table to identify the major specification differences offered by both brands.

<u>Router</u>

The table of comparison of router from two different brands namely the SG-5105, Multi-WAN Security Gateway and Huawei AR1220E in *Table 2* is as follows:

Specification/Product	FS SG-5105, Multi-WAN Security Gateway	Huawei AR1220E
Price(RM)	3,747.00	5,228.39
Performance	500 Mbps	400 Mbps
I/O Ports	9x 1 Gigabit, 1x 10 Gigabit WAN, 8x 1000BASE-T, 1x SFP, 1x SFP+	2 x GE Combo, 8 x GE (can be configured as WAN interfaces)
Memory	2 GB	1 GB
Flash	8 GB	512 MB

Table 2: FS SG-5105 vs Huawei AR1220E

The first major difference is the price range of these two routers. FS SG-5105 offers better specification than Huawei AR1220E even though Huawei AR1220E is more expensive. FS SG-5105 beats Huawei AR1220E in every single aspect such as performance for more reliable and stable connections, with less buffering and fewer dropped connections. It also has a more extensive list of I/O ports giving better connectivity options. Lastly, both memory and flash are better offered by FS SG-5105 compared to AR1220E.

Connectors

The table below, *Table 3* illustrates two different connectors which are the MP88X-1000 and the RS PRO Male RJ45 Connector for comparison to determine the best one for the project.

Feature	MP88X-1000	RS PRO Male RJ45 Connector
Price(RM)	1,101.56	1.231
Number of Positions/Contacts:	8p8c (RJ45, Ethernet)	8p8c (RJ45, Ethernet)
Shielding	Unshielded	Unshielded
Contact Finish	Gold	1 GB
Contact Material	Copper Alloy	Phosphor Bronze

Table 3 : MP88X-1000 vs RS PRO Male RJ45 Connector

From the table, both of them have almost the same specification while also not having a big price difference. The determining factor for this comparison is the contact material used by both connectors. From the table, MP88X-1000 uses copper alloys rather than phosphor bronze as its contact material. Copper alloy is significantly more superior than phosphor bronze in terms of its electrical conductivity for faster signal transfer. It is also malleable which is good to design a more flexible cable.

Fiber Optic Cable

Looking at the comparison table between Cisco SC-SC Single Mode Fiber Optic Cable and RS SC-SC Single Mode Fiber Optic Cable, several key differences in *Table 4* below:

Feature	Cisco SC-SC Single Mode Fiber Optic Cable	RS SC-SC Single Mode Fiber Optic Cable
Price(RM)	5.06/meter	54.51/meter
Maximum Insertion Loss	0.3dB	0.5dB
Return Loss	55dB	50dB
Fibre Type	OS2	OS1
Operating Temperature	-20℃to + 70℃	-20℃to + 60℃

Table 4 : Cisco SC-SC Single Mode Fiber Optic Cable vs RS SC-SC Single Mode Fiber Optic Cable

The most notable distinction lies in the pricing, where the RS cable is priced at 54.51 RM per meter, making it almost 11 times pricier than the Cisco cable, which costs 5.06 RM per meter. Other than that, the RS cable exhibits a slightly higher maximum insertion loss of 0.5dB, in contrast to Cisco's 0.3dB, indicating greater signal loss at connection points. The return loss attributes are also in favor of the Cisco cable, achieving 55dB as opposed to RS's 50dB, which suggests better control over signal reflection in the Cisco version. The types of fiber utilized also vary, with Cisco employing the more advanced OS2 standard, while RS relies on OS1, where OS2 generally provides superior performance over longer distances. Lastly, there is a minor difference in the operating temperature ranges; Cisco supports a broader range of -20°C to +70°C compared to RS’s -20°C to +60°C, giving an advantage of 10°C at the upper limit for the Cisco cable. In summary, the Cisco cable seems to offer enhanced specifications across most criteria while being significantly more economical than its RS equivalent.

Patch Panel

Looking at *Table 5*, let's break down the key differences between the Iwilink and Rapink 24-port Cat 7 patch panels:

Specification	Iwilink Patch Panel 24 Port Cat 7	Rapink Patch Panel 24 Port Cat 7
Price RM	223.47	236.88
Suitability	Rackmount, Cabinet, wallmount	Rackmount, Wallmount, Cabinet
Backward Compatibility	No	Yes
Connector Type	RJ45	RJ45

Table 5: Iwilink Patch Panel 24 Port Cat 7 vs Rapink Patch Panel 24 Port Cat 7

The Rapink model is slightly more expensive but offers backward compatibility, which is a significant advantage as it can work with older network equipment. If you have a mix of old and new network devices, the Rapink would be the better choice despite the higher price. However, if you're setting up a completely new network with all modern equipment, the Iwilink could save you some money since you won't need the backward compatibility feature.

Switch

The table below compares 2 available switches that are suitable for the labs:

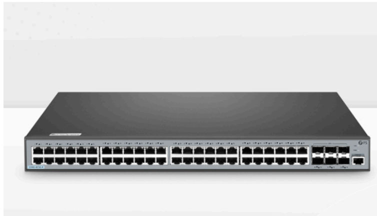
Product Name	X1052P - Dell Networking X Series Switches 	S3900-48T6S-R 
Ports	48	48
Switching capacity	176 Gbps	216Gbps
Forwarding rate	130.95 Mpps	162 Mpps
Power supply	100-240 VAC, 50/60 Hz	100-240 VAC, 50/60 Hz
Price	RM4,651.84	RM2,126.00

Table 6 : X1052P - Dell Networking X Series Switches vs S3900-48T6S-R

The Dell Networking X1052P switch and the S3900-48T6S-R switch offer similar 48-port configurations, yet differ significantly in performance and cost. The X1052P provides a switching capacity of 176 Gbps and a forwarding rate of 130.95 Mpps at RM4,651.84,whereas the S3900-48T6S-R delivers higher performance with a 216 Gbps switching capacity and 162 Mpps forwarding rate for a much lower price of RM2,126. For Ethernet cables, the CS67Z3 Category 7A S/FTP Cable features a delay of 25 ns/100m at RM14.59/m, while the LSZH Indoor IEC60332-3-24 Category 7A Enhanced Cable has a superior delay of 12 ns/100m at RM15.09/m. Thus, the S3900-48T6S-R and LSZH cable are cost-efficient choices offering higher performance.At the end,we choose S3900-48T6S-R as it offers higher performance at the suitable cost.

Ethernet cable

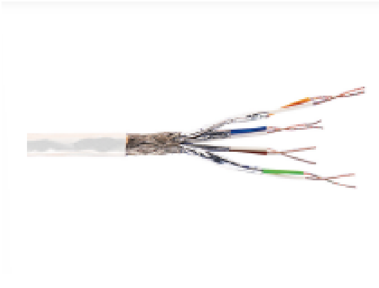
Product Name	CS67Z3 Category 7A S/FTP Cable 	LSZH Indoor IEC60332-3-24Category 7A Enhanced Cable 
ANSI/TIA Category	7A	7A
Delay	25 ns/100m	12ns/100m
Switching capacity	176 Gbps	216Gbps
Forwarding rate	130.95 Mpps	162 Mpps
Power supply	100-240 VAC, 50/60 Hz	100-240 VAC, 50/60 Hz
Price	RM 14.59 /m	RM 15.09 /m

Table 7 : CS67Z3 Category 7A S/FTP Cable vs LSZH Indoor IEC60332-3-24Category 7A Enhanced Cable

For the Ethernet cables, the CS67Z3 Category 7A S/FTP Cable and the LSZH Indoor IEC60332-3-24 Category 7A Enhanced Cable both meet the ANSI/TIA Category 7A standard, ensuring high-performance data transmission. However, the LSZH cable has a superior delay performance of 12 ns/100m compared to the CS67Z3's 25 ns/100m. The switching capacity and forwarding rate are linked to the respective switches they complement, with the LSZH cable supporting up to 216 Gbps and 162 Mpps, while the CS67Z3 supports up to 176 Gbps and 130.95 Mpps. In terms of cost, the CS67Z3 is slightly more affordable at RM14.59/m, while the LSZH cable is priced at RM15.09/m. The choice between the two depends on whether delay performance or cost is the priority.We choose the LSZH Indoor IEC60332-3-24 Category 7A Enhanced Cable as it offers lower delay and higher speed of forwarding rate.

Wireless Access Point

Specification	Fortinet FAP-431F 	Ubiquiti UAP-AC-PRO 
Price	RM 3,057.00	RM 665.00
Wi-Fi Standard	802.11ax (Wi-Fi 6)	802.11ac (Wi-Fi 5)
Maximum Data Transfer Rate	Radio 1 (2.4GHz): up to 1147 Mbps Radio 2 (5GHz): up to 2402 Mbps	Radio 1 (2.4GHz): up to 450 Mbps Radio 2 (5GHz): up to 1300 Mbps
Maximum Client	512 clients per radio (Radio 1 and Radio 2)	250 clients
Maximum Transmission Speed (Ethernet Port)	2500 Mbps	1000 Mbps
Operating Temperature	0°C to 45°C	-10 to 70° C

Table 8: Fortinet FAP-431 vs Ubiquiti UAP-AC-PRO

For the Wireless Access Point, FAP-431F is more expensive than the UAP-AC-PRO but the specifications are very different. FAP-431F supports Wi-Fi 6 which is an improved Wi-Fi version compared to UAP-AC-PRO which only supports Wi-Fi 5. The better Wi-Fi version improved the speed and connectivity of the wireless network. FAP-431F also supports more clients per radio compared to UAP-AC-PRO. FAP-431F has a higher transmission speed of 2500 Mbps compared to 1000 Mbps for the UAP-AC-PRO. FAP-431F operates with less temperature limit which will make it more suitable for extremely hot conditions.

Modem

Specification	Fortinet FortiWiFi-81F-2R-3G4G-POE 	NetGear CM3000 
Cellular Modem	3G4G / LTE	-
Ports	GE RJ45/SFP Shared Media Pairs x2, GE RJ45 POE/+ Ports x6, GE RJ45 POE/+ FortiLink Ports (Default) x2	LAN 2.5Gbps Gigabit Ethernet port with auto-sensing technology 1x, LAN 1Gbps Gigabit Ethernet ports with support for link aggregation 2x and WAN coaxial cable connection 1x
DOCSIS Compatibility	No	Yes
Firewall Supported	Yes	No

Table 9: Fortinet FortiWiFi-81F-2R-3G4G-POE vs NetGear CM3000

The FortiWiFi-81F is great for setups needing mobile network backup because it supports 3G/4G LTE. It has multiple wired connection options, including PoE ports for powering devices, and built-in firewall features to secure the network. However, it doesn't work with cable internet services like DOCSIS. This makes it ideal for businesses or complex networking environments. On the other hand, The NetGear CM3000 supports fast connections with its 2.5Gbps Ethernet port and link aggregation for multiple devices. However, it lacks a firewall. It's less suited for prioritizing network security. This modem is perfect for home or small office setups needing fast broadband without complex networking needs.

TASK 4: MAKING THE CONNECTIONS – LAN and WAN

4.1 IDENTIFY THE WORK AREAS ON FLOOR DIAGRAM

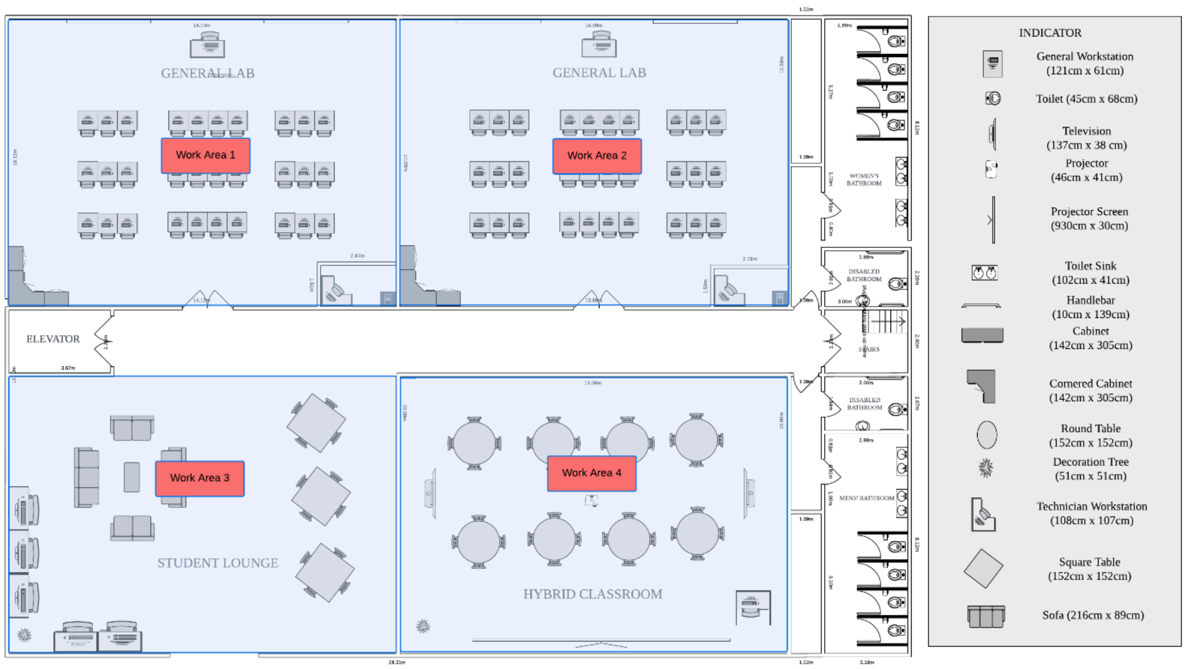


Figure 4.1 Work Areas on Ground Floor

As shown in Figure 4.1, Work Areas 1 and 2, are both general labs which has a size of 14 meters in length and 10 meters in width. Each work area 1 and 2 are intended for students during classes and it is equipped with 32 workstations, including those in the technician room. For each lab, it requires a 48-port switch and a single wireless access point that links all the workstations together.

Next, Work Area 3 in Figure 4.1 is a student lounge that is a refreshing area for students to complete their work, spanning 14 meters in length and 10 meters in width, with 5 working stations including a switch and wireless access point for those who are in dire need of using a computer. Onto the next part, Work Area 4 represents the hybrid classroom that is also 14 meters in length and 10 meters in width. This dynamic learning space consists of 8 roundtables that could fit up to 4 people per table, a projector and its panel, 2 TVs, and 1 pc totaling up to 3 workstations in this room also includes a switch and wireless access point.

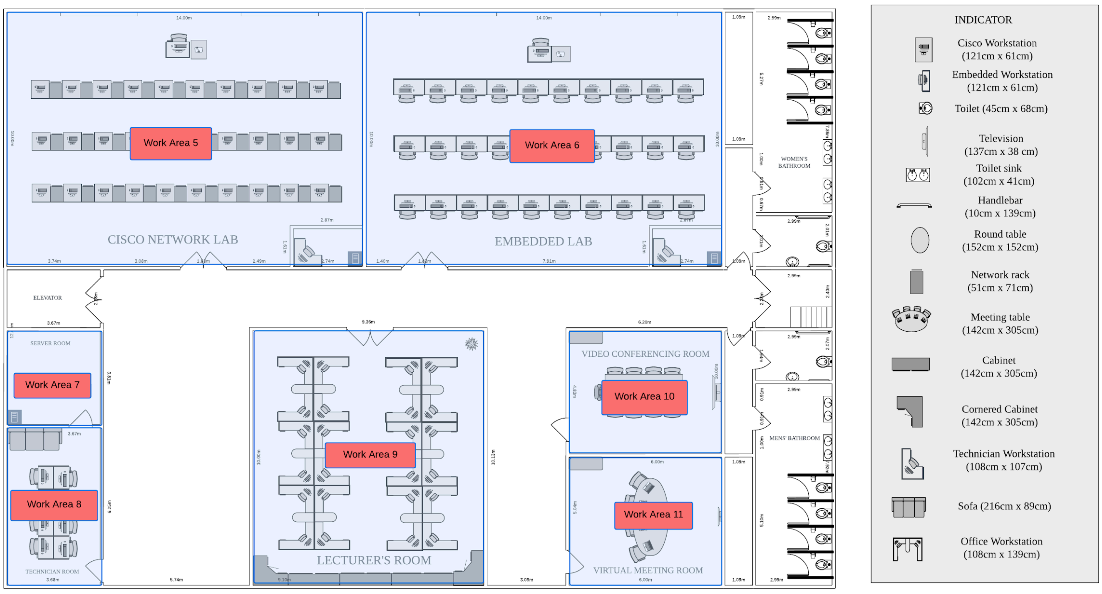


Figure 4.2 Work Areas at 1st Floor

Figure 4.2 illustrates the identified work areas on the 1st floor with a total of 7 work areas. Work area 5 and 6 consists of 32 workstations each where there are a total of 30 student's workstations, 1 lecturer's workstation and 1 technician's workstation. For each lab, it requires a 48-port switch and a single wireless access point that links all the workstations together.

Work area 7 is where the servers and telecommunications equipment is located. This is where the network is distributed to both floors. Next to it is the technician's room also known as work area 8 where only authorized technicians have access to the servers room to manage network and data services. The technician's room consists of 8 workstations.

Work area 9 is the lecturer's office where a total of 24 workstations are located. The room needs at least a 24-port switch to support the network distribution for all workstations. Finally, the video conferencing room and virtual meeting room is identified as work area 10 and 11. Both of these rooms do not require as extensive as other rooms as it only has 2 devices that supports internet connection which is a smart television.

4.2 THE DISTRIBUTION AND CONNECTION DEVICES

4.2.1 NETWORK DISTRIBUTION DIAGRAM

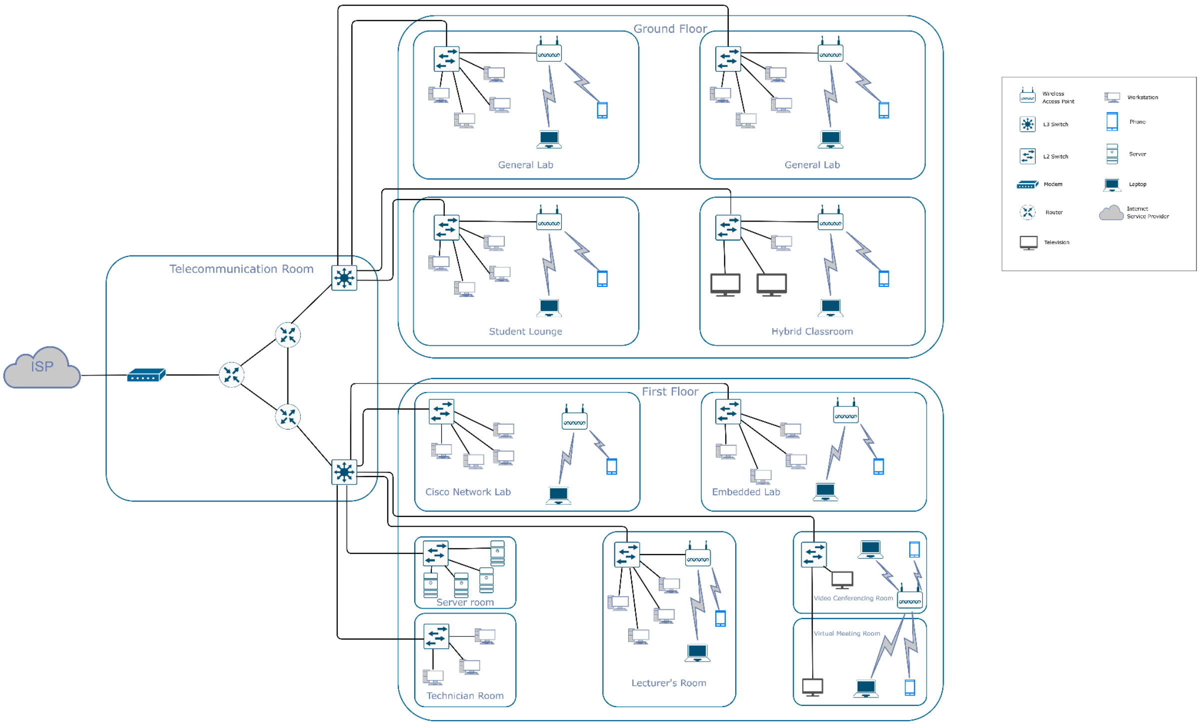


Figure 4.3 Network Distribution Diagram

The Network Distribution diagram illustrates the connection of the network with different rooms which is focussed on how good the connectivity of the internet is established. Figure 4.3 shows the Network Distribution diagram for the ground floor and first floor. Each floor shows the connectivity to each lab room.

This Network Distribution diagram split to three layers which are core layer, distribution layer and access layer. The core layer is the layer that covers the modem and router. The distribution layer covers the two L3 switches that distribute connectivity to each floor. The access layer covers the switch and wireless access point that make connections for all the devices in each room.

For a clearer figure of the network distribution on the ground and first floors, refer the figure 4.4 and figure 4.5.

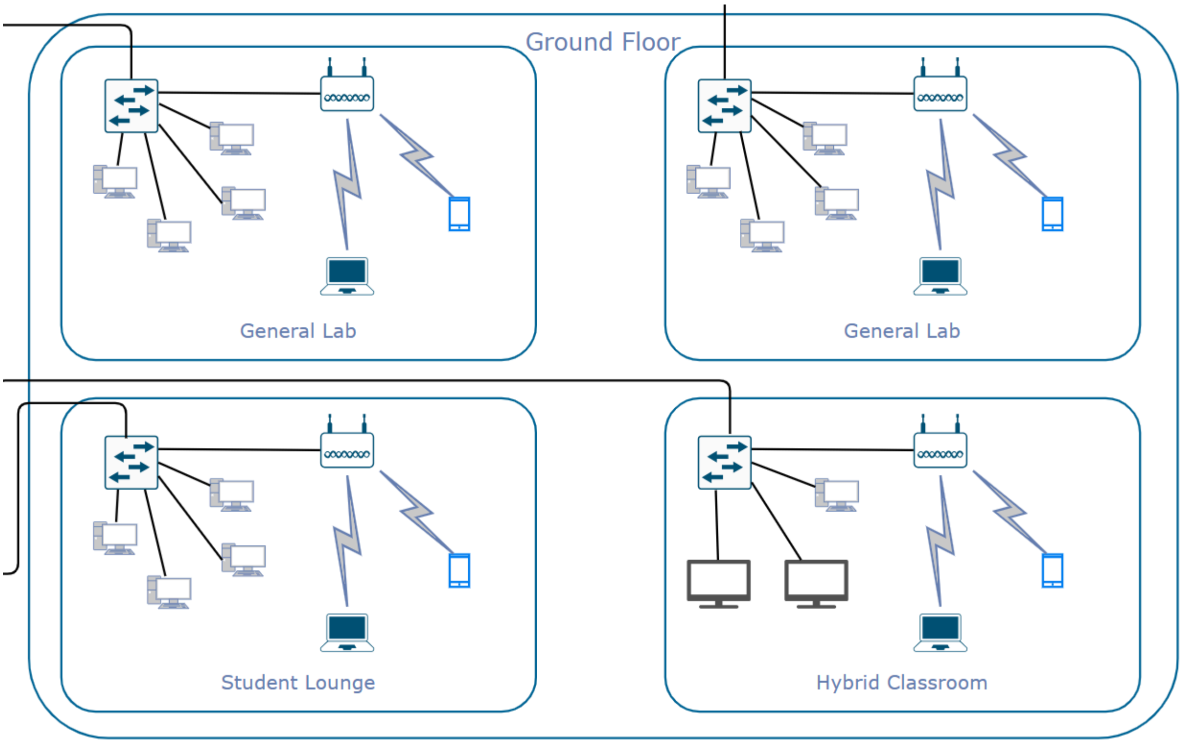


Figure 4.4 Network Distribution Diagram for Ground Floor

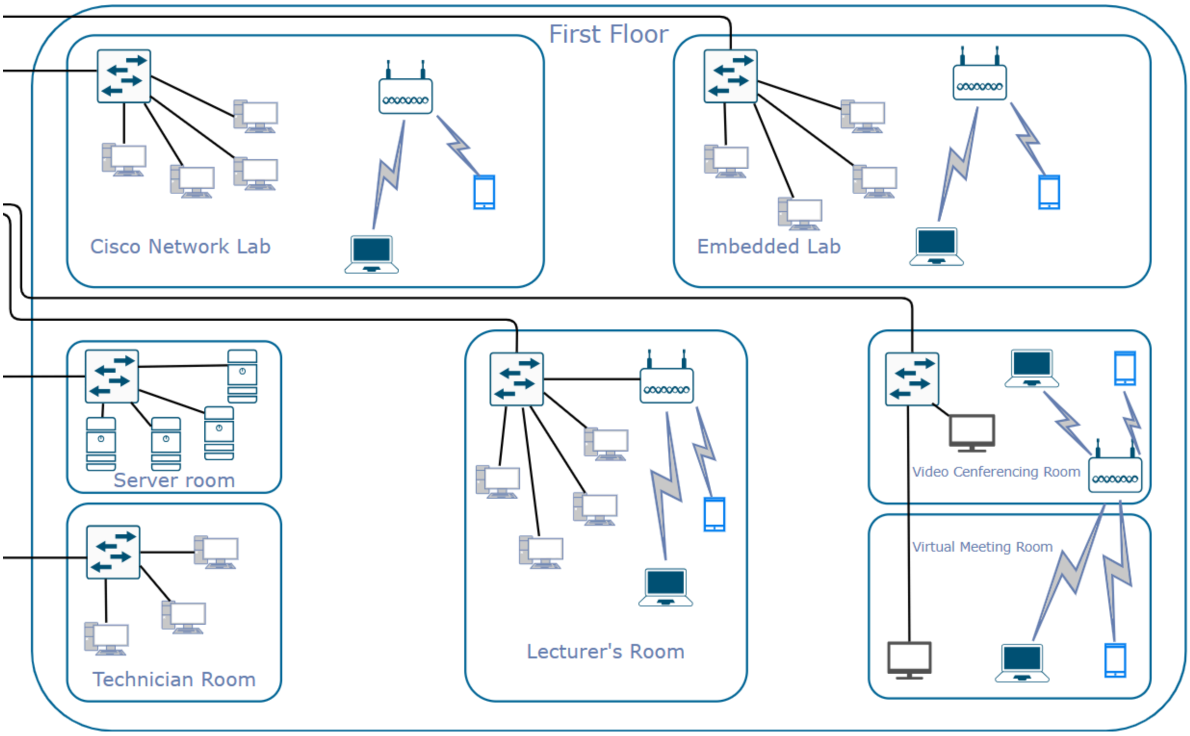


Figure 4.5 Network Distribution Diagram for First Floor

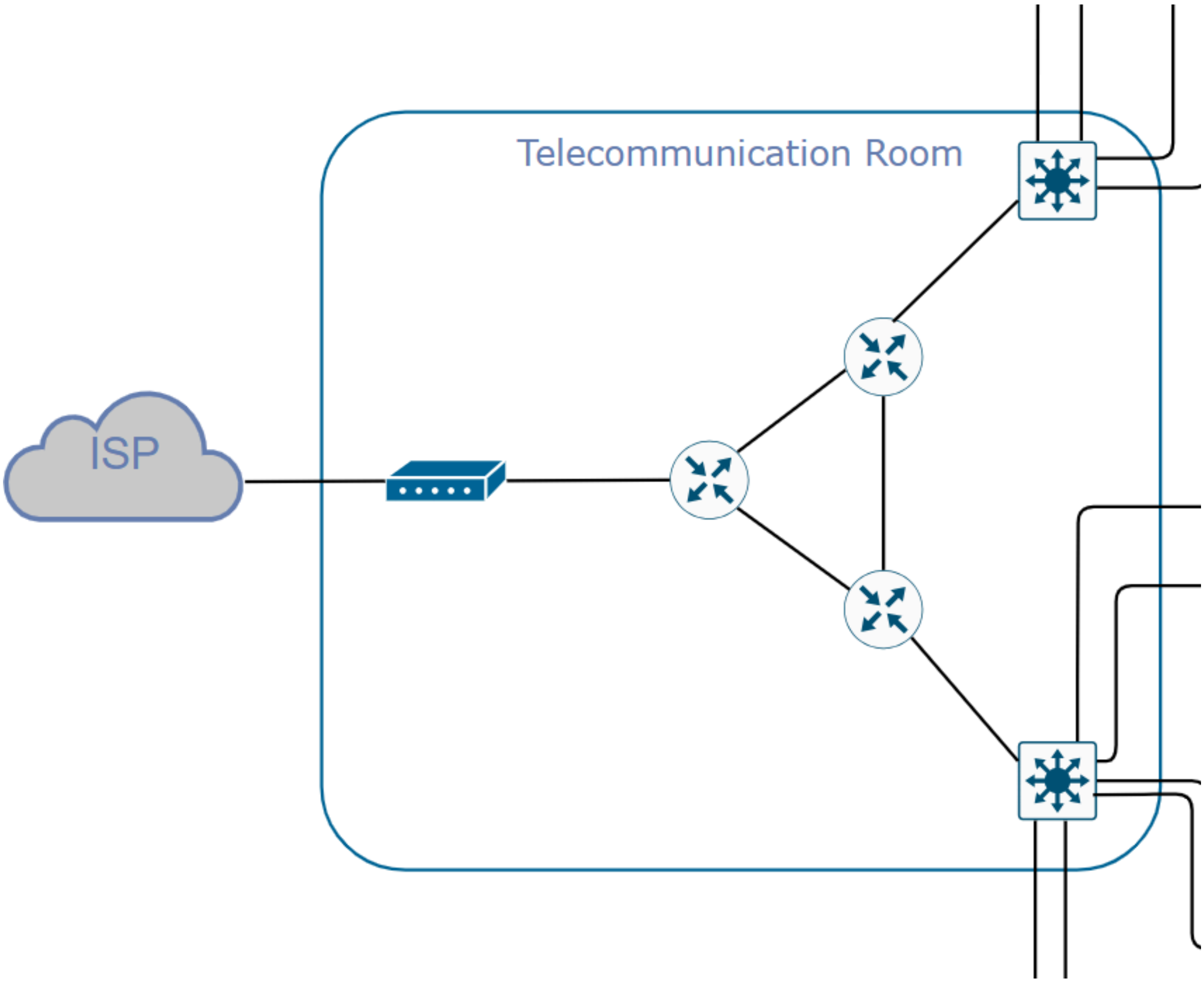


Figure 4.6 Network Distribution Diagram - Telecommunication Room

Figure 4.6 shows the network distribution within the telecommunication room which is focused on the core and distribution layer. These layers have routers, modem and L3 switch. In this setup, the modem is connected by the ISP and then all the routers. Routers that used many instead of one because it will enhance the connectivity performance. L3 switch connected from the router and then connected to all switches for each floor. The location for the telecommunication room is inside the server room for the server room easier to be connected to the server.

4.2.3 FLOOR DIAGRAM

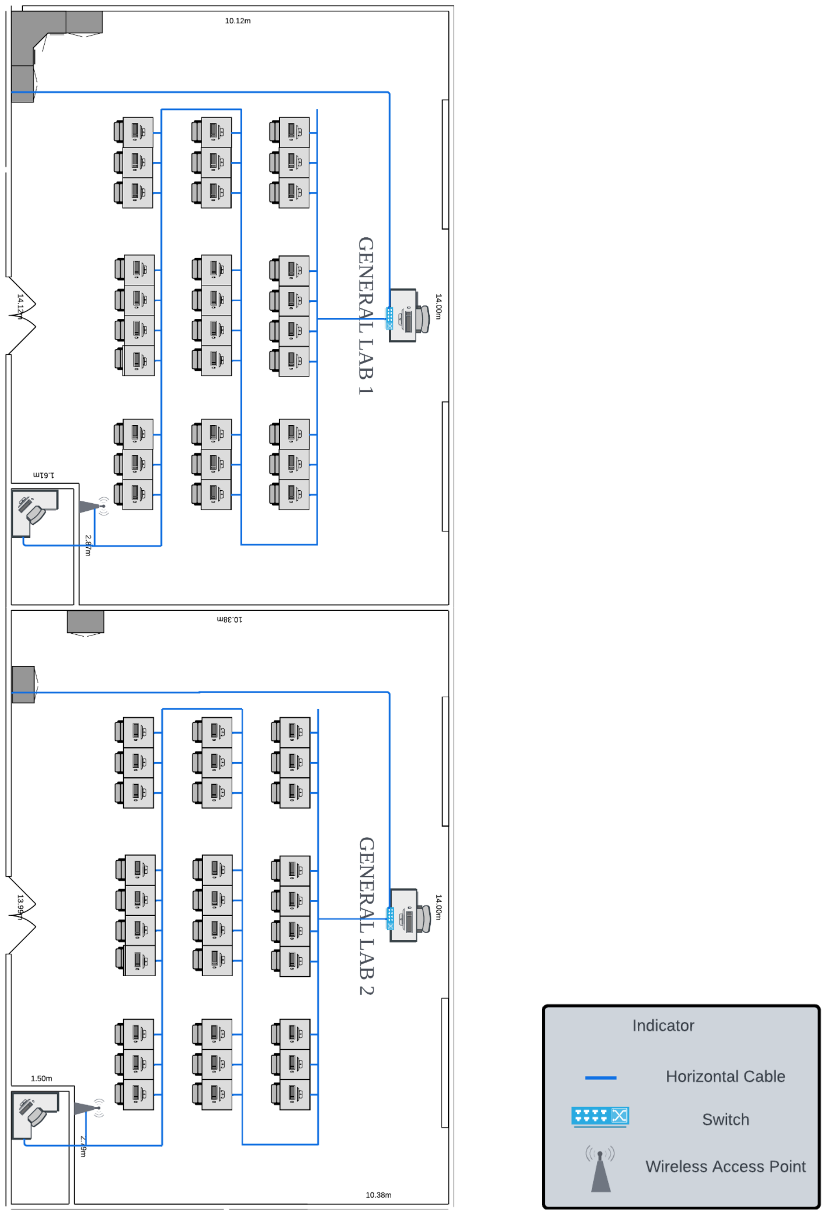


Figure 4.7 Connections in General Labs

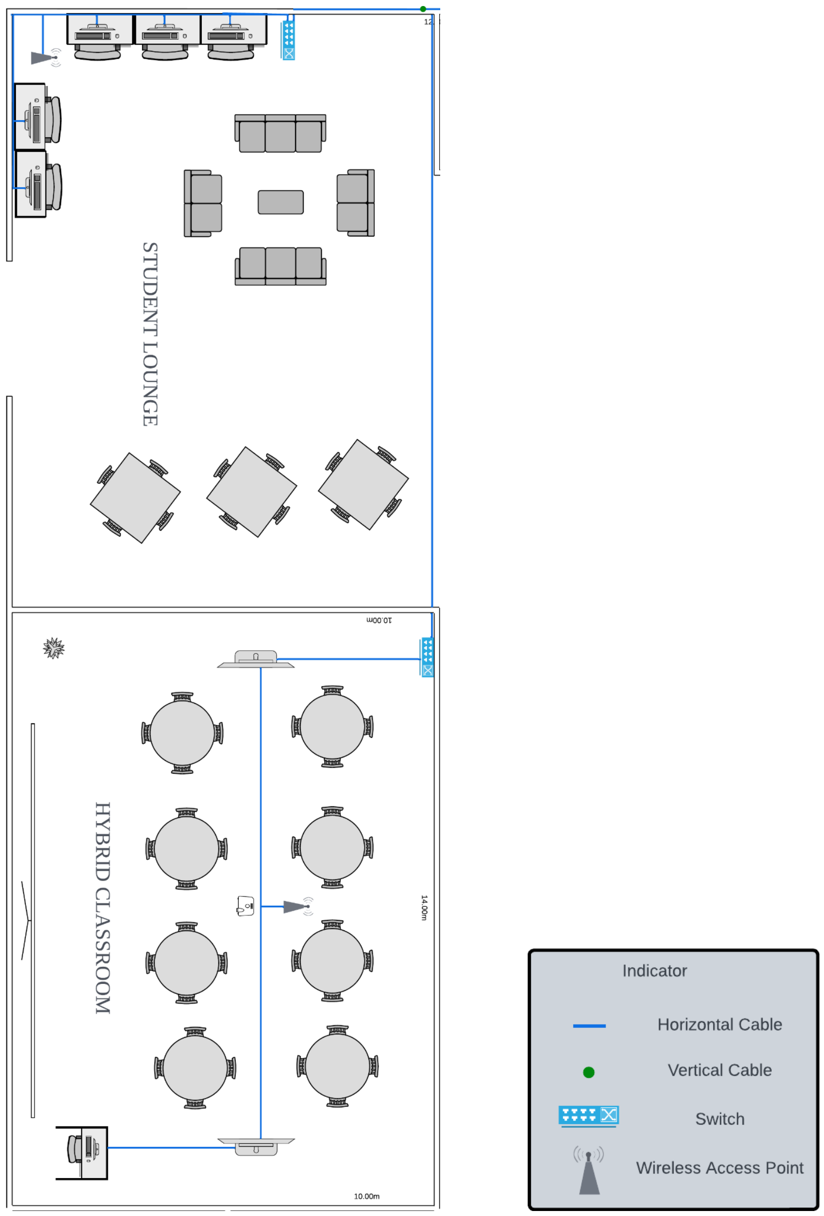


Figure 4.8 Connections in Hybrid Classroom and Student Lounge

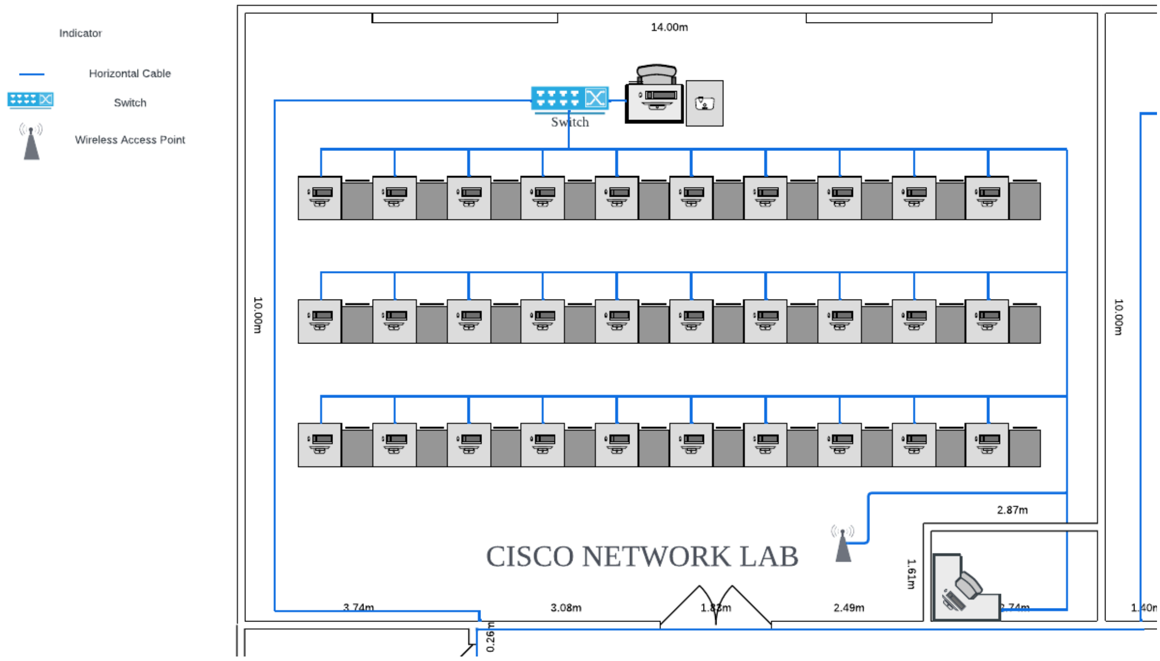


Figure 4.9 Connections in Cisco Networking Lab

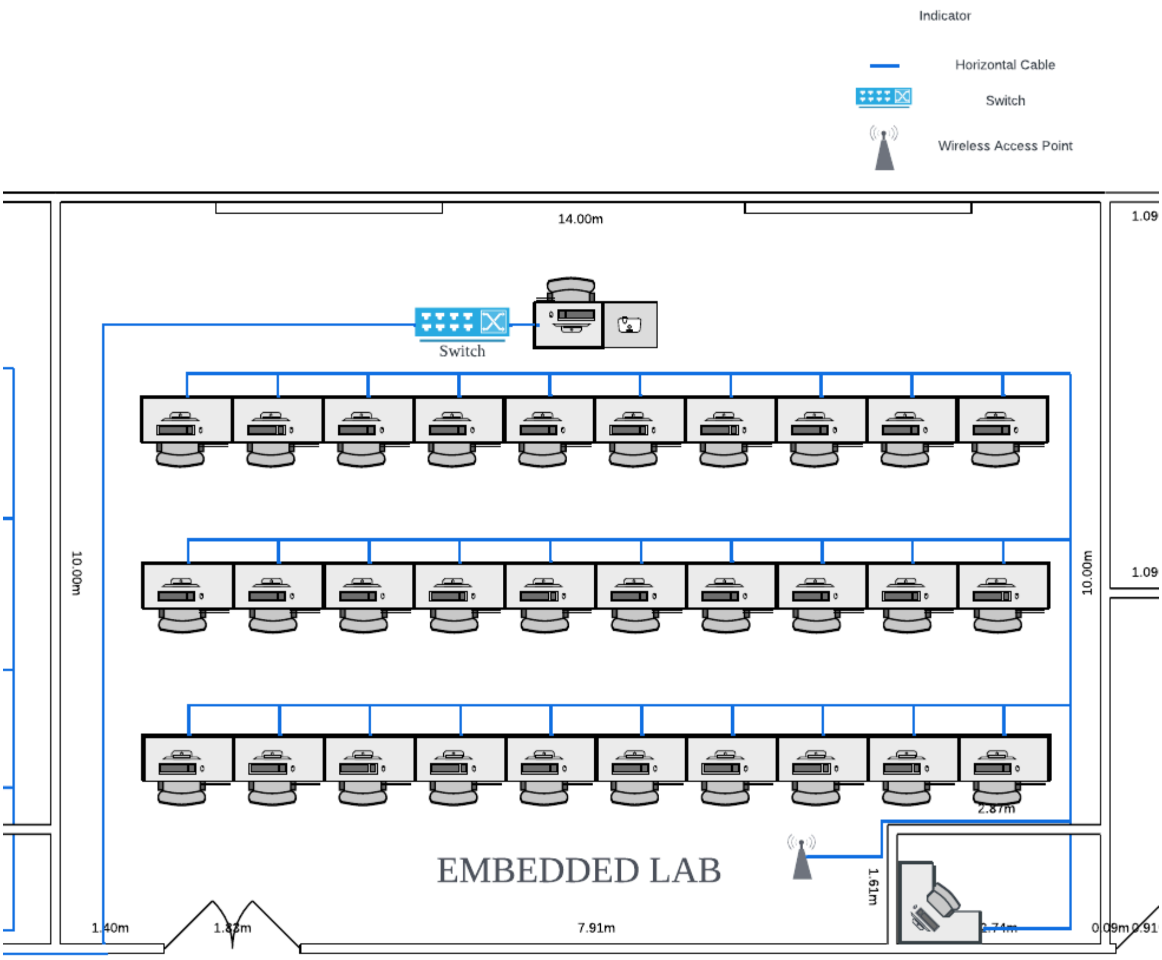


Figure 4.10 Connections in Embedded Lab

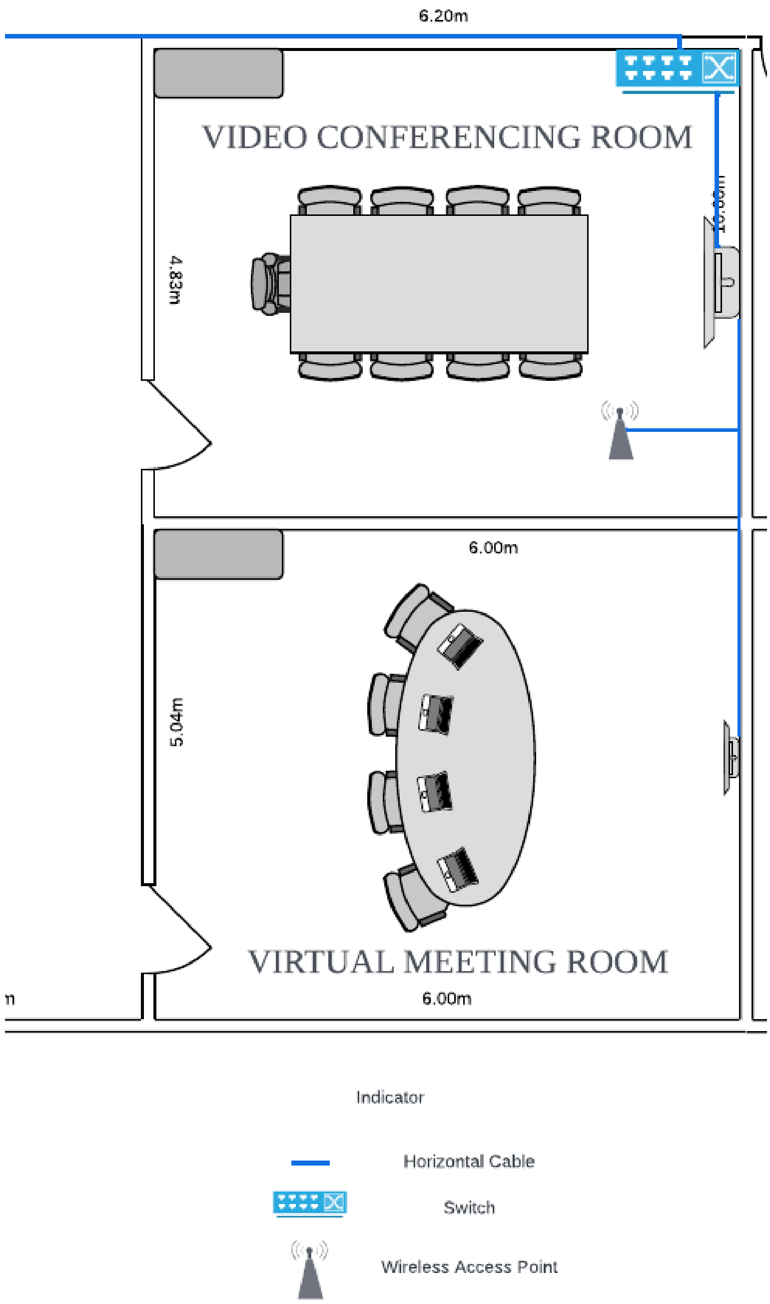
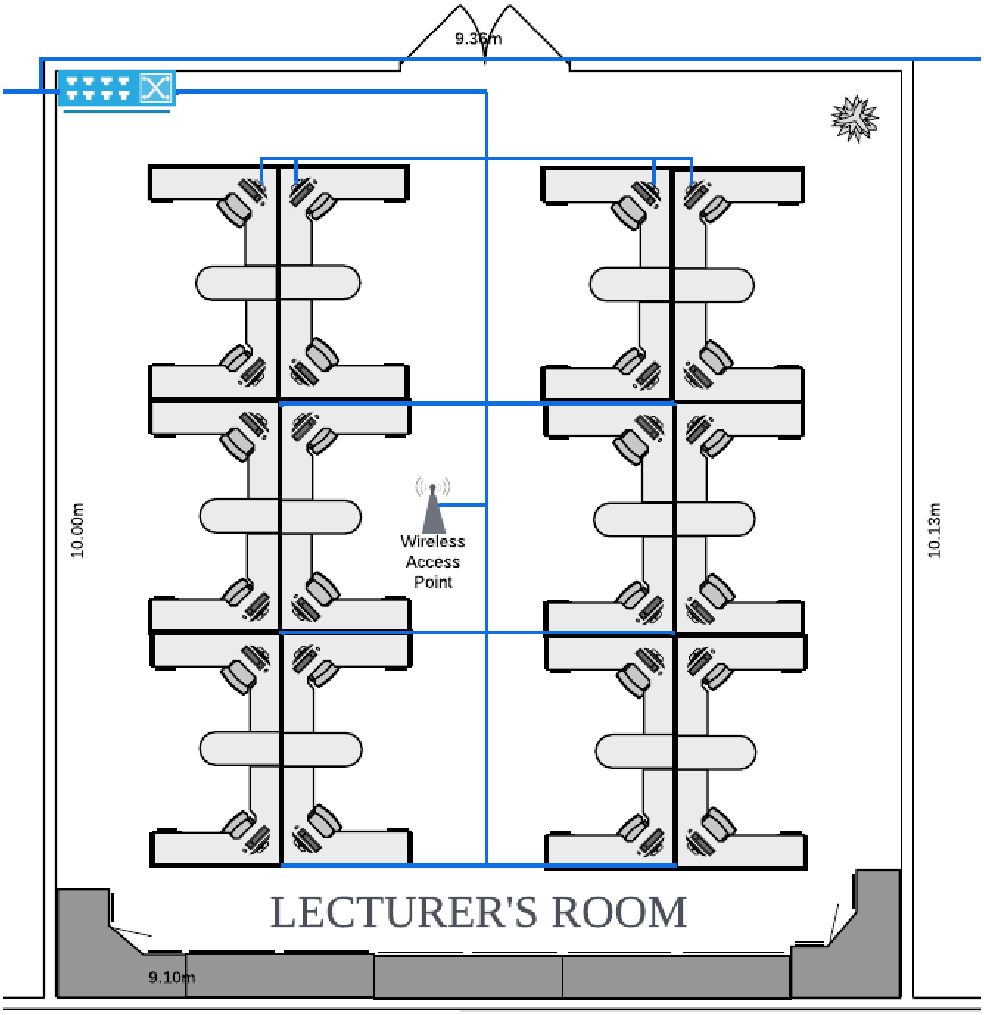
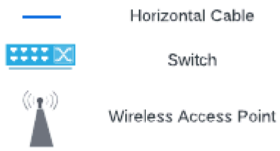


Figure 4.11 Connections in Video Conferencing Room and Virtual Meeting Room


9.36m
10.00m
10.13m
Wireless
Access
Point
LECTURER'S ROOM
9.10m

Indicator


Horizontal Cable
Switch
Wireless Access Point

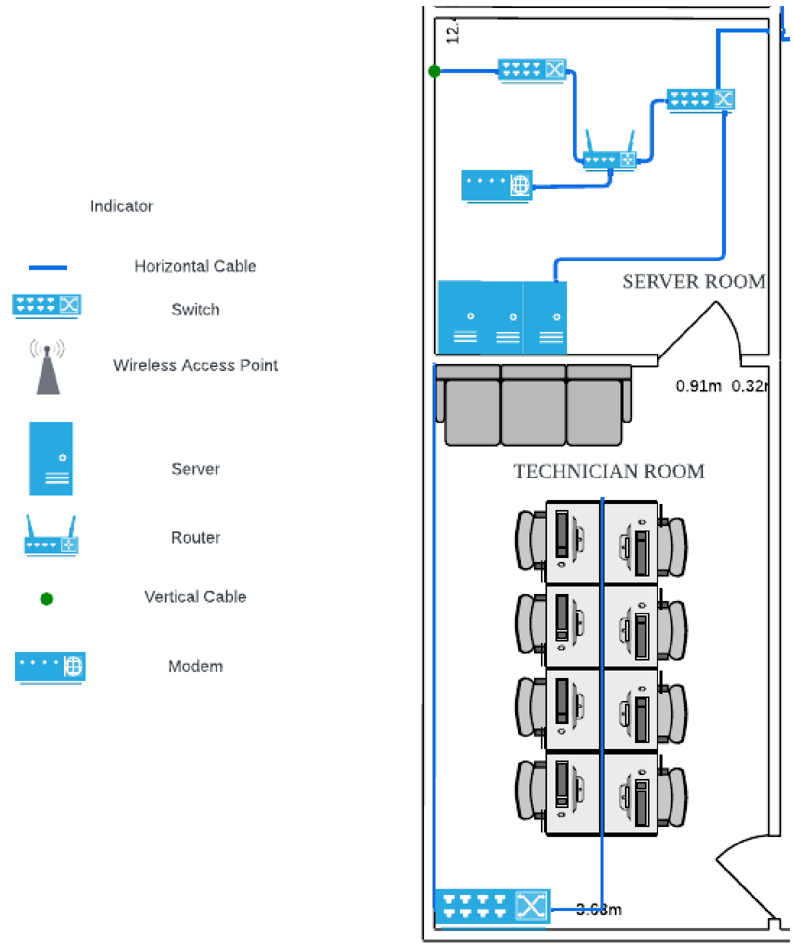


Figure 4.12 Connections in Server Room and Technician's Room

4.3 NUMBER OF CONNECTIONS, PATCH CORDS AND SWITCH PORTS NEEDED

Area	Number of Switch Ports Needed
General Lab 1	33
General Lab 2	33
Hybrid Classroom	5
CISCO Network Lab	33
Embedded Lab	33
Video Conferencing Room	2
Virtual Meeting Room	1
Lecturer’s Room	25
Server Room	1
Technician Room	9
Student Lounge	6
Total	181

Table 4.3.1: Number of Switch Ports Needed

Area	Number of Patch Cords Needed
General Lab 1	33
General Lab 2	33
Hybrid Classroom	5
CISCO Network Lab	33
Embedded Lab	33
Video Conferencing Room	2
Virtual Meeting Room	1
Lecturer’s Room	25
Server Room	1
Technician Room	9
Student Lounge	6
Total	181

Table 4.3.2: Number of Patch Cords Needed

Area	Number of RJ45 Connector Needed
General Lab 1	66
General Lab 2	66
Hybrid Classroom	10
CISCO Network Lab	66
Embedded Lab	66
Video Conferencing Room	4
Virtual Meeting Room	2
Lecturer’s Room	50
Server Room	2
Technician Room	18
Student Lounge	12
Total	362

Table 4.3.3: Number of RJ45 Connector Needed

4.4 IDENTIFY CABLE TYPES AND LENGTH

Area	Length (m)		Total Length (m)	Total Price (RM)
	Horizontal	Vertical		
Ground Floor				
General Lab 1	255	0	255	3847.95
General Lab 2	255	0	255	3847.95
Hybrid Classroom	98	0	98	1478.82
Student Lounge	66	3	69	1041.21
Ground Floor Total			674	10,215.93

Area	Length (m)		Total Length (m)	Total Price (RM)
	Horizontal	Vertical		
First Floor				
CISCO Network Lab	280	0	280	4225.2
Embedded Lab	280	0	280	4225.2
Video Conferencing Room	10	0	10	150.9
Virtual Meeting Room	10	0	10	150.9
Lecturer's Room	194	0	194	2927.46
Server Room	20	0	20	301.80
Technician Room	35	0	35	528.15
First Floor Total			829	12,509.61

Info about the cable used:
LSZH Indoor IEC60332-3-24Category 7A Enhanced Cable -RM 15.09/m

Ground Floor		
Path	Length(m)	Total Price (RM)
Server room to Student Lounge	16	80.90
Server room to Hybrid Classroom	19	96.06
Server room to General Lab 1	25	126.40
Server room to General Lab 2	42	212.35
TOTAL	102	515.71

First Floor		
Path	Length(m)	Total Price (RM)
Server room to Technician Room	16	80.90
Server room to Lecturer's Room	11	55.62
Server room to Video Conferencing Room	31	156.74
Server room to CISCO Network Lab	28	141.57
Server room to Embedded Lab	37	187.07
TOTAL	123	621.90

Info about the cable used:
SC-SC-10-Meter-Singlemode-Fiber-Optic-Cable-*RM 5.06/m*

Two types of cables have been deployed to ensure seamless connectivity across the building: LSZH Indoor IEC60332-3-24 Category 7A Enhanced Cable and SC-SC Singlemode Fiber Optic Cable. These cables were strategically chosen based on their performance and suitability for the student in the Faculty of Computing's needs.

Category 7A Enhanced Cables are utilized extensively for horizontal and vertical cabling within both the ground and first floors. This cable type supports high-speed data transmission with minimal interference, making it ideal for connecting devices in general labs, hybrid classrooms, lounges, and offices. On the ground floor, these cables link key areas such as General Lab 1, General Lab 2, the hybrid classroom, and the student lounge to the network. Similarly, on the first floor, the same cabling connects the CISCO Network Lab, Embedded Lab, lecturer’s room, and other spaces to the central network system.

SC-SC Singlemode Fiber Optic Cables serve as the backbone for long-distance and high-bandwidth connections, ensuring fast and efficient data transfer between the server room and critical areas on both floors. On the ground floor, fiber optic cables create connections from the server room to General Lab 1, General Lab 2, the hybrid classroom, and the student lounge. On the first floor, they extend connectivity from the server room to the CISCO Network Lab, Embedded Lab, video conferencing room, lecturer’s room, and technician room.

By combining the strengths of Category 7A Enhanced Cables for internal connections and fiber optic cables for inter-floor and backbone connections, the network is optimized for reliability, speed, and efficiency, meeting the building's advanced communication and data transfer needs.The strategic placement of these cables ensures that each area of the building is connected efficiently, supporting both high-speed internal communication and seamless data transfer to external networks. The Category 7A Enhanced Cables are ideal for typical network applications within rooms like labs and classrooms, where high data throughput is needed for devices such as computers, projectors, and printers. Meanwhile, fiber optic cables are crucial for interconnecting critical infrastructure, such as the server room allowing for high-capacity data flow over long distances without signal degradation. This combination guarantees robust internal connectivity, while also providing reliable, fast links to the Internet and other external services, ensuring the building's network can handle large volumes of data traffic with minimal downtime.

TASK 5: IP ADDRESSING SCHEME

5.1 NETWORK ADDRESS GIVEN BY LECTURER

Group	Network Address
1	10.16.0.0/12
2	10.32.0.0/12
3	10.48.0.0/12
4	10.64.0.0/12
5	10.80.0.0/12
6	10.96.0.0/12
7	10.112.0.0/12
8	10.128.0.0/12
9	10.144.0.0/12
10	10.176.0.0/12

Figure 5.1.1

Figure 5.1.1 presents the list of network addresses assigned to each group. Our group's network address is 10.16.0.0/12.

Network Address details:

IP Address(decimal)	10.16.0.0
IP Address(binary)	00001010 00010000 00000000 00000000
Subnet Mask(decimal)	255.240.0.0
Subnet Mask(binary)	11111111 11110000 00000000 00000000

The subnet mask is partitioned into a network portion (16 bits) and a host portion (16 bits). After performing the AND operation of the IP address with the subnet mask, we identified the network address as 10.16.0.0 and the broadcast address as 10.16.255.255. From that, we know that the valid address range is between 10.16.0.1 and 10.16.255.254.

Network Address	10.16.0.0
Broadcast Address	10.31.255.255
Range of usable address	10.16.0.1 - 10.31.255.254

5.2 THE BEST POSSIBLE WAY TO DIVIDE THE NETWORK

5.2.1 THE BEST POSSIBLE WAY TO DIVIDE FOR THE NETWORK

Using classful addressing, we divided the network into subnets by floor level and then only by room based on the work areas defined in Task 4. The ground floor will use the IP address of the 10.16.0.0 and the first floor will use the IP address of 10.17.0.0.

Subnet	Network Portion	Host Portion		IP Address
0	00001010 00010000	00000000 00000000	Network Address	10.16.0.0
		11111111 11111111	Broadcast Address	10.16.255.255
1	00001010 00010001	00000000 00000000	Network Address	10.17.0.0
		11111111 11111111	Broadcast Address	10.17.255.255

The following table below uses the IP address of 10.16.0.0 with the subnetting mask of 24

Subnet	Network Portion	Host Portion		IP Address
2	00001000 00010000 00000000	00000000	Network Address	10.16.0.0
		11111111	Broadcast Address	10.16.0.255
3	00001000 00010000 00000001	00000000	Network Address	10.16.1.0
		11111111	Broadcast Address	10.16.1.255
4	00001000 00010000 00000010	00000000	Network Address	10.16.2.0
		11111111	Broadcast Address	10.16.2.255
5	00001000 00010000 00000011	00000000	Network Address	10.16.3.0

		11111111	Broadcast Address	10.16.3.255

The following table below uses the IP address of 10.17.0.0 with the subnetting mask of 24

Subnet	Network Portion	Host Portion		IP Address
6	00001000 00010001 00000000	00000000	Network Address	10.17.0.0
		11111111	Broadcast Address	10.17.0.255
7	00001000 00010001 00000001	00000000	Network Address	10.17.1.0
		11111111	Broadcast Address	10.17.1.255
8	00001000 00010001 00000010	00000000	Network Address	10.17.2.0
		11111111	Broadcast Address	10.17.2.255
9	00001000 00010001 00000011	00000000	Network Address	10.17.3.0
		11111111	Broadcast Address	10.17.3.255
10	00001000 00010001 00000100	00000000	Network Address	10.17.4.0
		11111111	Broadcast Address	10.17.4.255
11	00001000 00010001 00000101	00000000	Network Address	10.17.5.0
		11111111	Broadcast Address	10.17.5.255
12	00001000 00010001 00000110	00000000	Network Address	10.17.6.0
		11111111	Broadcast Address	10.17.6.255

Distribution of subnet for different work areas:

Subnet	Work Area	Network Address	Broadcast Address	Range of usable address	Subnet Mask Address
Ground Floor (subnet 0)					
2	General Lab	10.16.0.0	10.16.0.255	10.16.0.1 - 10.16.0.254	/24 255.255.255.0
3	General Lab	10.16.1.0	10.16.1.255	10.16.1.1 - 10.16.1.254	/24 255.255.255.0
4	Student Lounge	10.16.2.0	10.16.2.255	10.16.2.1 - 10.16.2.254	/24 255.255.255.0
5	Hybrid Classroom	10.16.3.0	10.16.3.255	10.16.3.1 - 10.16.3.254	/24 255.255.255.0
First Floor (subnet 1)					
6	Cisco Network Lab	10.17.0.0	10.17.0.255	10.17.0.1- 10.17.0.254	/24 255.255.255.0
7	Embedded Lab	10.17.1.0	10.17.1.255	10.17.1.1 - 10.17.1.254	/24 255.255.255.0
8	Video Conferencing Room	10.17.2.0	10.17.2.255	10.17.2.1 - 10.17.2.254	/24 255.255.255.0
9	Virtual Meeting Room	10.17.3.0	10.17.3.255	10.17.3.1 - 10.17.3.254	/24 255.255.255.0
10	Lecturer's Room	10.17.4.0	10.17.4.255	10.17.4.1 - 10.17.4.254	/24 255.255.255.0
11	Server Room	10.17.5.0	10.17.5.255	10.17.5.1 -	/24

				10.17.5.254	255.255.255.0
12	Technician Room	10.17.6.0	10.17.6.255	10.17.6.1 - 10.17.6.254	/24 255.255.255.0

CONCLUSION

Working on this project as a group taught us how important teamwork is. We learned how to collaborate by using everyone's commitment to complete tasks. Clear communication and listening were important to avoid misunderstandings and solve any issues. We also realized that managing time well is very important. When things didn't go as planned, like making mistakes with the project requirements, we learned to adapt and make changes quickly. Sharing ideas with each other helped us improve our problem-solving skills, and we understood how important it is for everyone to do their part well for the success of the team.

When doing the project, we discovered that designing floor plans is harder than we think. We learned that there are many tools available to help with this design. We learned that good research is important to meet project requirements and prevent mistakes like mistaken the original project requirements. We also gained a huge understanding of how network devices work.We also realize that setting up a network for even one building is quite expensive and complicated. This makes us appreciate the network systems in our faculty. Additionally, we improved our knowledge of subnetting and learned how to divide a network address for a department.

This experience taught us valuable things like teamwork, planning, and problem solving skills. It is really important to make sure this project completes and meets its requirements.

TEAM MEMBERS AND RESPONSIBILITIES

No	Member	Responsibility
1.	Iman Abadi Bin Mohd Nizwan	For my contributions in Task 1, I designed the Cisco Networking Lab and the Embedded Lab by identifying the proper layouts and equipment to meet the requirements of those rooms. In Task 2, the contribution I made was answering three questions that were generated by our team. The questions answered were “What networking devices are required for the Cisco Network Lab to support teaching, and how will they align with future industry needs?”, “How can the network be optimized to support cloud-based applications and services that the Faculty of Computing may adopt in the future?” and “What type of cable is suitable to set up a networking environment among computers based on the specification?”. In Task 3, I was assigned to research routers for the Cisco Networking Lab and the faculty, connectors and switches for the Cisco Networking Lab. I picked the most suitable one for our project while also keeping in mind the scalability requirements of the project. In Task 4, I was responsible for designing the connections of the 1st floor of our plan. From this task, I have learned how all of the devices researched in Task 3 are interconnected and their respective roles when designing a network. Finally in Task 5, all of us were involved in researching how subnetting is done when designing a network. Choosing the proper subnet mask, identifying the usable host addresses to accommodate all of the devices in the faculty, and dividing the subnets the best ways possible were all learnt in this task.
2.	Mohamed Alif Fathi Bin Abdul Latif	For task 1, my contribution is I made a dummy sketch of the ground and first plan. Then, we decided to divide the task by each member designing one to two rooms. My part is to design video conferencing rooms and virtual meeting rooms. After design, we compile each room. For task 2, I made questions 2, 4 and 6. For task 3, my part is searching and explaining the wireless access point and modem for a table of devices. Then, I answered question 3 for reflection on my part, which is major differences between the same devices from different brands for modem and wireless access point. For task 4, my contribution is

		to complete the network diagram section. I designed the network diagram for each floor and compiled it. For task 5, I suggest the subnet mask for every room and each floor. I also make the calculation for every subnet mask and the network address detail which is to find the network address, usable address and the broadcast address for every subnet mask. For task 6, my part is to define the table of figures and copy the part from task 5 to the compiled task. I also make myself part of team and member responsibilities.
3.	Afif Shaqir Irfan bin Arqam	In this project, I helped the group by researching suitable software for Task 1 and selected Lucidchart for designing the floor plan due to its real-time cloud collaboration features. I also designed the Hybrid Classroom and General Lab, researching and suggesting suitable equipment like projectors and TVs to ensure functionality. Additionally, I created a feasibility assessment based on technical, economic, and operational aspects in Task 2 and generated one question and researched the technical and spatial requirements for the Hybrid Classroom using articles and journals from Google Scholar. In task 3,I ensured tasks were divided equally among group members according to rubrics and suggested switches to support the network requirements of all labs.I manage to choose suitable numbers of switch port needed with the lab and room requirements. I also researched and sourced cost-effective ethernet cables, surfed for high-quality switches, and identified the cable types and lengths. I determined the number of connections, patch cords, and switch ports by collaborating on distribution diagrams and calculating totals for each room in Task 4. Lastly in Task 5, I actively contributed to planning the IP addressing scheme, optimizing subnetting for labs, classrooms, and offices while minimizing address wastage. My efforts ensured the network

		design was logical, practical, and met both academic and practical requirements.
4.	Muhammad Adam bin Razali	In each of the tasks, I had key responsibilities that were very vital for the success of the project. These are summarized below: In Task 1, I was able to design the ground floor, including the general lab and toilet, propose the design of rooms and equipment in them. In Task 2, I probed some important questions to be used in assessing the needs of the network architecture, especially on its security, monitoring, and disaster recovery. For Task 3, I selected fiber optic cables and patch panels for use in the building and estimated their number and cost. In Task 4, I developed the strategy for the routing of cables at the ground floor, ensuring that each and every workplace was connected to a single point that connects to the server room. Lastly, in Task 5, the entire network was designed for IP addressing, and each member was tasked with the calculation of the subnet mask that would fit each workplace. In so doing, the network design will be optimized, and all is executed within the budget.

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APPENDICES

FINANCIAL BUDGET

Item	Price per unit (RM)	Quantity	Total Price (RM)
Router	3,747.00	3	11,241.00
Access Point	3,057.00	8	24,456.00
Modem	18,670.00	1	18,670.00
Connector	2.42	362	876.04
Switch	2,126.00	9	19,134.00
Patch Panel	223.47	20	4,469.40
Cisco Networking Lab Router	4,461.57	8	35,692.56
Cisco Networking Lab Switch	3,275.21	8	29,801.68
Fibre Optic Cable	5.06/m	225m	1,138.50
Patch Cord	15.09/m	1,503m	22,680.27
		Total	168,159.45

The total amount of expenses based on the table above is **RM 168,159.45**. The budget assigned to our group (Team 1) is **RM 3,000,000.00**. The amount that is left after deduction is **RM 2,831,840.55**. The total planned expenses is just **5.6%** of our budget.

MEETING MINUTE TASK 1

Meeting Minute #1

Date/Time	30 October 2024 8PM		
Location	Afif's Room		
Agenda	- Discuss the primary task of the project - Identify software needed to build the faculty plan - Divide the task		
Meeting Host	Iman Abadi		
Attendance			
Name	**Time**	**Reason For Absence**	
Iman Abadi	2000	-	
Alif Fathi	2000	-	
Muhd Adam	2000	-	
Afif Shaqir	2000	-	
Minutes			
No.	**Item Discussed**	**Ideas/Suggestion and Person Giving It**	**Person in Charge & Date**
1	Group name selection	- Afif suggested the group name is "AlphaTech" but was rejected due to it being used by another group project of another subject. - Alif suggested the group name is "InNet" and everybody approved.	Alif **30/10/2024**
2	Case study discussion	- Everyone read the case study individually before it was discussed. - Discussion was done by having each	Iman **30/10/2024**

		member list the requirements of the study. - Afif asked suggestions on whether the rooms with no dimension specification have to follow any requirements. Iman answered that the specification is up to us.	
3	Software to be use	- Alif suggested using EdrawMax however the app is not free. - Afif suggested using LucidChart software because it is free. - We proceeded with choosing LucidChart as our designing software.	Afif **30/10/2024**
4	Splitting workload	- Adam facilitated the workload division by asking volunteers to take up on what room they want to design. - Iman volunteered to design the Cisco Networking Lab and the Embedded Lab - Adam volunteered on designing 2 general labs. - Alif volunteered on designing the virtual meeting room and the video conferencing room. - Afif volunteered to design the hybrid classroom. - Adam Reminds everyone to recheck on the rubric and objective of the task	Adam **30/10/2024**

		to make sure nothing crucial were left out before executing the floor design	
5	Next meeting	- Alif suggested that everyone should be done with their work by 2/11/2024 and everyone approved	All members **30/10/2024**
6	Meeting Ended	2300	All members **30/10/2024**

Meeting Minute #2

Date/Time	2 November 2024		
Location	Afif's Room		
Agenda	- Constructing floor plan for the project		
Meeting Host	Iman Abadi		
Attendance			
Name	**Time**	**Reason For Absence**	
Iman Abadi	1500	-	
Alif Fathi	1500	-	
Muhd Adam	1500	-	
Afif Shaqir	1500	-	
Minutes			
No.	**Item Discussed**	**Ideas/Suggestion and Person Giving It**	**Person in Charge & Date**
1	Floor plan construction	- All members contribute in suggesting the layout of the building based on their design specification.	All Members **2/11/2024**
2	Lecturer's Room	- Everyone forgot to allocate a room for the lecturers as everyone thought that we only need to provide space for an increasing number of students, but then Afif reminded everyone that the academic staffs are also increasing so a room for lecturers is needed.	Afif **30/10/2024**
3	Hybrid Classroom and General Lab	- Research and suggesting suitable equipment to be put	Afif **2/11/2024**

		inside of the classroom to make sure it functions perfectly as it should such as projector and tv - Planning the classroom design to be fit for 40 students at a time(not including lecturer). - Making sure the general lab has all it needs to run a massive amount of desktop computers with ease including an ample and long-lasting desk that is cable-through friendly.	Adam **2/11/2024**
4	Cisco Networking Lab and Embedded Lab	- Iman verified the proper layout for all the rooms assigned to him including the tools needed for the room based on a short search from google to get the rough idea of the room structure and proposed it to be on the 1st floor because of his plan of having the technician's office to also be on the 1st floor.	Iman **2/11/2024**
5	Video Conferencing Room	- Alif designed the conferencing room with all requirements for the room included. - The video conferencing room has been designed	Alif **2/11/2024**

		with a complete networking connection and two big televisions to access the virtual meeting with ease.	
6	Other rooms and facilities	- Alif designed the elevator and stairs in the plan and approved by other team members. - Alif designed the toilet to be in the middle of the building but was rejected by Iman. - Adam suggested the toilet to be built at the right side of the building and approved by team members. - Alif suggested Server Room to be on the ground floor but rejected because the server room was better on the first floor due to potential floods that might happen. - Iman designed the technician's office and suggested it be placed on the 1st floor as the server room, Cisco Networking Lab and Embedded Lab is also located there. This is because these rooms require more attention to help troubleshoot any issues. - Afif suggested the student lounge area to be in front of the	All Members **2/11/2024**

		building and in an open air space.the student lounge area also provide study place and discussions chair for the student and academic staff use.	
7	Final adjustments	- All members made sure that the floor plan satisfies the case study's requirements.	All members **2/11/2024**
8	Meeting ended	- Meeting ended at 23.00 After all the work had been done. - The report was completed.	All Members **2/11/2024**

MEETING MINUTE TASK 2

Meeting minute #1

Date/Time	6 November 2024 4.30 PM
Location	Arked Angkasa
Agenda	- Discuss the primary task of the project - Divide the task
Meeting Host	Iman Abadi

Attendance		
Name	**Time**	**Reason For Absence**
Iman Abadi	1630	-
Alif Fathi	1630	-
Muhd Adam	1630	-
Afif Shaqir	1630	-

Minutes			
No.	**Item Discussed**	**Ideas/Suggestions and Person Giving It**	**Person in Charge & Date**
1	Case study Task 2 discussion	- Everyone read the case study individually before it was discussed. - The discussion was done by having each member list the requirements of the study.	Adam 6/11/2024
2	Question proposal	- Everyone proposed questions related to the case study - Each member suggested around 2-3 questions	Alif 6/11/2024
3	Filter questions	- Only the top 10 related questions were selected for this	Iman 6/11/2024

		task. - Iman suggested using the remaining questions related to the task as a backup if any of the 10 questions cannot be answered.	
4	Task distribution	- Iman suggested that 3 members each take 3 questions, one member takes 1 question and one does the feasibility study. - Iman, Alif and Adam are assigned with 3 questions each - Afif is assigned one question and the feasibility study	Afif 6/11/2024
5	Meeting ended	- The meeting ended at 1800.	

Meeting Minute #2

Date/Time	8 November 2024 4.00 PM
Location	Afif's Room
Agenda	- Review answers from the questions assigned - Answer corrections if there are any - Discuss the feasibility study based on the questions answered
Meeting Host	Iman Abadi

Attendance

Name	Time	Reason For Absence
Iman Abadi	1600	-
Alif Fathi	1600	-
Muhd Adam	1600	-
Afif Shaqir	1600	-

Minutes

No.	Item Discussed	Ideas/Suggestions and Person Giving It	Person in Charge & Date
1	Explain the answer to each question	- Each presented and explained their answer.	Iman 8/11/2024
2	Review the answer	- Iman suggested making corrections if the question and answer were out of topic. - Corrections are determined by votes from other members.	Alif 8/11/2024
3	Answer correction	- All questions that needed corrections were done together. - Each member was assigned to find research materials related to the question.	Adam 8/11/2024

		- The questions were answered together based on the reference material researched.	
4	Feasibility assessment	- Afif made the feasibility assessment based on the questions answered. - Afif asked if the feasibility assessment was in the correct format.	Afif 8/11/2024
5		- Adam suggested that the assessment needs to be based on the questions that were answered. - Alif suggested that the feasibility needs to be done based on three aspects which are operational, technical and economic feasibility.	
6	Meeting ended	- The meeting ended at 2030	All members

MEETING MINUTE TASK 3

Meeting Minute #1

Date/Time	14 October 2024 8PM
Location	Dewan Sri Resak
Agenda	- Discussion about tasks requirement - Divide task for every member
Meeting Host	Iman Abadi

Attendance

Name	Time	Reason For Absence
Iman Abadi	2000	-
Alif Fathi	2000	-
Muhd Adam	2000	-
Afif Shaqir	2000	-

Minutes

No.	Item Discussed	Ideas/Suggestions and Person Giving It	Person in Charge & Date
1	Task 3 requirements discussions	-Alif asked the group members to read the task instructions. -Adam and Iman discussed and planned the budget to complete the task. -Afif makes sure that the tasks are divided equally	ALL

		among group members and follows the task's rubrics to avoid any problems	
2	Networking Devices Used	- Alif suggested that the campus should have wireless access points for every lab and modem for connecting the ISP through fiber optic cables. - Adam suggested that the campus should have enough cables to cover all networks around the campus. - Afif suggest that the campus should have switches that will support the network requirements on all lab that are going to be built. - Iman suggest the campus should have routers for routing the network.	ALL
	Meeting ended	- At 11.30pm	All

Meeting Minute #2

Date/Time	16 October 2024 8PM	
Location	Afif's Room	
Agenda	- Discuss	
Meeting Host	Iman Abadi	
Attendance		
Name	**Time**	**Reason For Absence**
Iman Abadi	2000	-
Alif Fathi	2000	-
Muhd Adam	2000	-
Afif Shaqir	2000	-

Minutes

No.	**Item Discussed**	**Ideas/Suggestion and Person Giving It**	**Person in Charge & Date**
1	Amount of the devices	Alif suggested that X number of wireless access points be put in every lab and area that requires a high-speed internet connection	
2	Determine the best network devices	- Iman remind us that patch panel as well as firewall are optional devices for the LAN and server	**All**

		is also not necessary yet to be put. - All of us recheck whether all of devices from the chosen brand could sustain future changes and growth in terms of, number of user and changes in modern technology - Alif Fathi draw us the network infrastructure/architecture for this network plan which emphasize on the proper position and numbers of the network devices such as modem, router and switches for the network plan.	
	Finalize the total budget and device needed	- All of us calculate total price for our respective devices and sum up - We also check our devices are within our budget given for	**All**

		the network plan	
	Meeting ended	- At 12am	

MEETING MINUTE TASK 4

Meeting Minute #1

Date/Time	13 November 2024 4.30 PM
Location	MA-1 KTDI
Agenda	- Discuss the primary task of the project - Divide the task
Meeting Host	Iman Abadi

Attendance

Name	Time	Reason For Absence
Iman Abadi	1630	-
Alif Fathi	1630	-
Muhd Adam	1630	-
Afif Shaqir	1630	-

Minutes

No.	Item Discussed	Ideas/Suggestions and Person Giving It	Person in Charge & Date
1	Case study Task 4 discussion	- Everyone read the case study individually before it was discussed. - The discussion was done by having each member list the requirements of the study.	Adam 13/12/2024
2	Suggestion on network distribution	- Iman suggested that for each rooms, certain network devices should be present in order to distribute network properly - Discussion on what network devices should be in each	Alif 13/12/2024

		room was researched/	
3	Task distribution	- Alif was assigned to design a network distribution diagram based on the discussion on network distribution. - Iman was assigned to design the network connections on the 1st floor of the building - Adam was assigned to design the network connections on the ground floor of the building. - Afif was assigned to calculation based on the cable length needed based on the network connections done.	Iman 13/12/2024
4	Meeting ended	- The meeting ended at 1800.	

Meeting Minutes #2

Date/Time	16 November 2024 2.00 PM		
Location	MA-1 KTDI		
Agenda	- Discuss the primary task of the project - Divide the task		
Meeting Host	Iman Abadi		
Attendance			
Name	**Time**	**Reason For Absence**	
Iman Abadi	1400	-	
Alif Fathi	1400	-	
Muhd Adam	1400	-	
Afif Shaqir	1400	-	
Minutes			
No.	**Item Discussed**	**Ideas/Suggestions and Person Giving It**	**Person in Charge & Date**
1	Review each members work	- Alif's work on the distribution diagram was reviewed by all members to check whether all the devices included in the floor plans are present. - Iman and Adam's work was reviewed by identifying if all connections to related work stations are all made. - Afif's work was reviewed by identifying the calculations based on the connection design from the floor plan.	Alif 16/12/2024

2	Making corrections on the cable length identified	- Some mistakes were identified in this part. - All members take part in rechecking the calculations for each room of the floor plan.	Afif 16/12/2024
3	Finalizing the report	- Each member took part in finalizing the report.	Iman 16/12/2024
4	Meeting ended	- The meeting ended at 1800.	

MEETING MINUTE TASK 5

Meeting Minutes #1

Date/Time	19 December 2024 11.30 AM
Location	MA-1 KTDI
Agenda	- Discuss the primary task of the project - Completing the report
Meeting Host	Iman Abadi

Attendance

Name	Time	Reason For Absence
Iman Abadi	1130	-
Alif Fathi	1130	-
Muhd Adam	1130	-
Afif Shaqir	1130	-

Minutes

No.	Item Discussed	Ideas/Suggestions and Person Giving It	Person in Charge & Date
1	Case study Task 5 discussion	- Everyone read the case study individually before it was discussed. - The discussion was done by having each member list the requirements of the study.	Adam 19/12/2024
2	Suggestion on subnetting	- Every member offers the subnetwork division that is suitable. - After discussion, each member chooses the option that best suits the idea.	Alif 19/12/2024
3	IP Addressing	- Every member	Iman

	Scheme Overview	checks to see how each work area's IP address is currently assigned.	19/12/2024
4	Proceed to the parts of Task 5	- All members work on google docs together to complete task 5. - While completing the tasks, each member checks each other's works if there are any errors and corrects them promptly.	Afif 19/12/2024
5	Meeting ended	- The meeting ended at 1600.	